Internship Report on

Adaptive UKF and Neural Modelling Based GPS-Denied Navigation Algorithm for UAV

Carried out at

Flight Mechanics Control Division

by

ADAM NABEEL (1CR22EC007)

under the guidance of

Dr.Majeed Mohamed

Senior Principal Scientist

Flight Mechanics Control Division

CSIR-National Aerospace Laboratories

Bengaluru, Karnataka - 560017

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ABSTRACT

Accurate and reliable state estimation is essential for autonomous Unmanned Aerial Vehicles (UAVs), particularly in environments where Global Positioning System (GPS) signals are weak or unavailable. Under nominal conditions, the integration of the Inertial Navigation System (INS) with GPS using an Adaptive Unscented Kalman Filter (AUKF) provides dependable navigation performance. However, during GPS outages, the system relies solely on INS measurements, which leads to rapid error accumulation due to sensor noise, bias, and integration drift. This limitation significantly affects the accuracy and stability of UAV navigation in real-world scenarios.

To address the challenges of navigation in GPS-denied environments, this study proposes a hybrid Adaptive Unscented Kalman Filter–Neural Network (AUKF–NN) framework for robust navigation and state estimation. The proposed approach initially employs an AUKF-based INS/GPS integrated navigation system to generate accurate reference state estimates during periods of GPS availability. These reference states are subsequently utilized to train a feedforward Neural Network capable of learning the nonlinear error characteristics of standalone INS navigation.During GPS outages, the trained Neural Network predicts the INS estimation errors and generates pseudo-GPS measurements. These measurements are then incorporated into the UKF update process to mitigate error accumulation and compensate for INS drift.

The proposed framework is validated using real-world flight datasets, demonstrating significant improvements in position and velocity estimation accuracy compared with standalone INS operation.The results further indicate enhanced navigation stability and reduced error growth during prolonged GPS-denied periods. Although the achieved performance does not completely match that of continuous GPS-aided navigation, the proposed AUKF–NN framework provides a reliable and effective solution for maintaining navigation accuracy in challenging GPS-denied environments.

Keywords— Unmanned Aerial Vehicles (UAV), Adaptive unscented Kalman Filter (AUKF), Inertial Navigation System (INS), GPS-denied navigation, Neural Network (NN), Sensor Fusion, Pseudo-GPS

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ABBREVIATIONS

AUKF	Adaptive Unscented Kalman Filter
EKF	Extended Kalman Filter
UKF	Unscented Kalman Filter
GPS	Global Positioning System
GNSS	Global Navigation Satellite System
INS	Inertial Navigation System
IMU	Inertial Measurement Unit
NN	Neural Network
MLP	Multi-Layer Perceptron
NED	North-East-Down
UAV	Unmanned Aerial Vehicle
MSE	Mean Squared Error
ReLU	Rectified Linear Unit

CHAPTER 1

Introduction

Unmanned Aerial Vehicles (UAVs) have gained significant importance in modern aerospace applications such as surveillance, environmental monitoring, disaster management, precision agriculture, and autonomous transportation. Reliable navigation is one of the most critical requirements for UAV operation, as accurate estimation of position, velocity, and attitude directly affects flight stability and mission performance. In most practical systems, navigation is achieved through the integration of the Global Positioning System (GPS) and Inertial Navigation System (INS), where GPS provides long-term positioning accuracy while INS offers high-rate motion estimation using inertial sensors. The fusion of INS and GPS enables continuous and robust navigation under normal operating conditions.

Despite these advantages, conventional INS/GPS integrated navigation systems face significant challenges in GPS-denied environments. UAVs operating in urban canyons, dense forests, tunnels, indoor environments, or regions affected by signal jamming may experience partial or complete GPS signal loss. During such conditions, the navigation system relies entirely on INS-based propagation, where accelerometer and gyroscope measurements are continuously integrated to estimate UAV states. However, sensor noise, bias, and modeling uncertainties accumulate over time, resulting in unbounded drift in position and velocity estimates. This degradation significantly reduces navigation accuracy and affects the reliability of autonomous UAV operation.

To address these challenges, this work proposes a novel hybrid state estimation framework that combines the Adaptive Unscented Kalman Filter (AUKF) with a Neural Network (NN) for robust UAV navigation in GPS-denied environments. The AUKF provides accurate nonlinear state estimation and sensor fusion capabilities, while the Neural Network learns the nonlinear error characteristics of the INS and predicts correction terms during GPS outages. By integrating machine learning with model-based estimation, the proposed approach improves navigation continuity, reduces INS drift, and enhances estimation accuracy under uncertain operating conditions.

1.1 GPS Velocity Measurements

To understand the navigation characteristics of the UAV, GPS velocity measurements obtained from the flight dataset are first analyzed. These measurements provide reference velocity information used by the navigation filter during normal GPS availability.

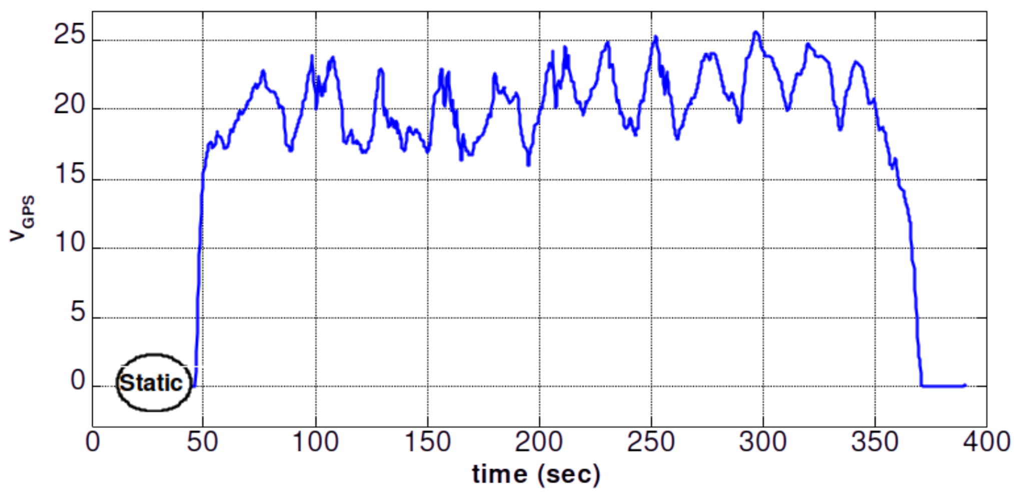

Figure 1.1.1: GPS-based velocity components in the NED frame

The North, East, and Down velocity components shown in Figure 1.1.1 represent the translational motion of the UAV in the NED coordinate frame. These velocity measurements serve as important correction inputs to the navigation filter during normal GPS availability. The variations observed in the velocity profiles reflect changes in UAV maneuvering and flight dynamics. During GPS outages, such measurements become unavailable, forcing the navigation system to rely solely on inertial propagation and resulting in increasing estimation errors.

1.2 GPS-Denied Navigation

The proposed navigation framework integrates the Adaptive Unscented Kalman Filter (AUKF) and Neural Network (NN) to achieve accurate UAV state estimation under both normal and GPS-denied conditions. During normal operation, the IMU provides accelerometer and gyroscope measurements, while GPS supplies position information.

These measurements are processed by the AUKF, which performs recursive prediction and correction to estimate UAV states such as position, velocity, and attitude. The AUKF continuously reduces sensor uncertainty and improves overall navigation accuracy.

During GPS outages, the correction capability of the AUKF becomes limited due to the absence of external measurements, leading to INS drift accumulation. To overcome this limitation, the Neural Network is integrated into the framework to learn the nonlinear error characteristics of the INS using previously available sensor and state data. The trained NN predicts correction terms, which are supplied to the AUKF as pseudo-measurements during GPS-denied operation. This hybrid AUKF–Neural Network approach enhances navigation continuity, minimizes estimation drift, and improves robustness in uncertain environments.

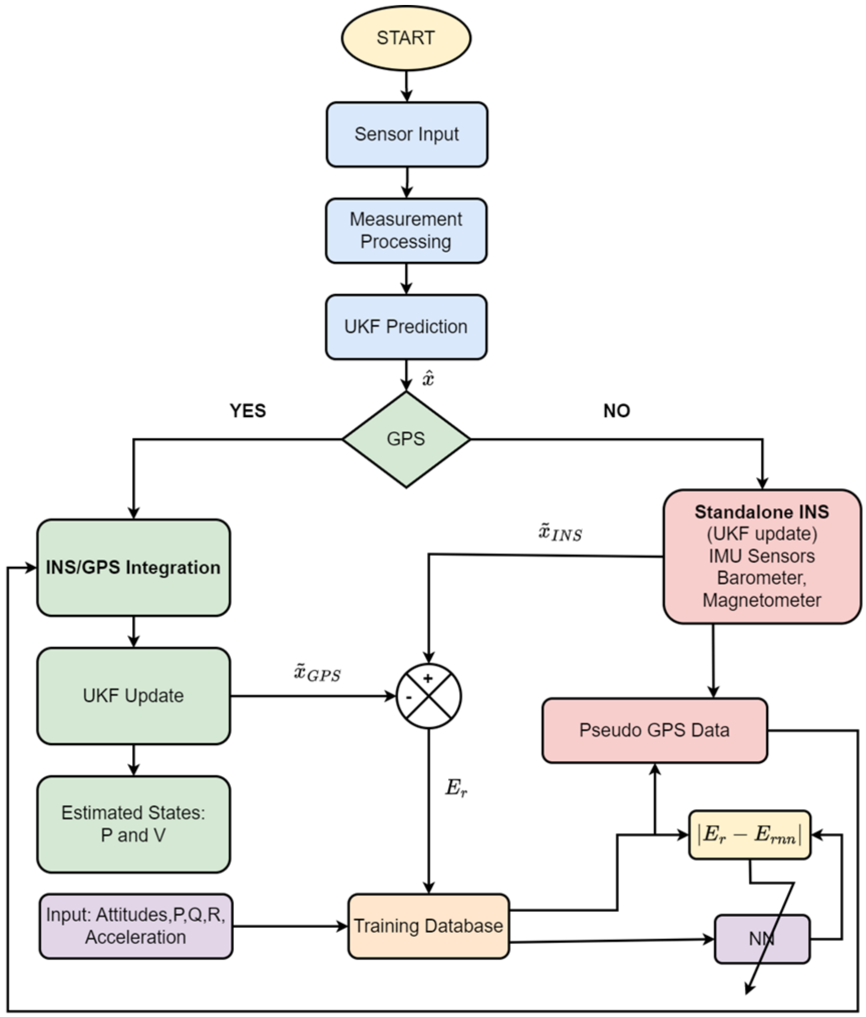

Figure 1.2.1: Architecture of the proposed AUKF–Neural Network based GPS-denied navigation framework

The remainder of this report is organized as follows. Chapter II presents the INS/GPS integrated navigation framework and sensor fusion methodology used for UAV state estimation. Chapter III discusses nonlinear filtering algorithms with emphasis on the Adaptive Unscented Kalman Filter (AUKF). Chapter IV describes the Neural Network architecture and its role in learning INS error characteristics during GPS-denied operation. Chapter V presents the state estimation results, simulation analysis, and performance evaluation of the proposed framework. Finally, Chapter VI summarizes the conclusions and outlines future research directions.

CHAPTER 2

INS/GPS Integrated Navigation

The INS/GPS integrated navigation framework combines the advantages of inertial sensing and satellite-based positioning to achieve accurate and continuous UAV state estimation. The Inertial Navigation System (INS) provides high-rate motion information using accelerometers and gyroscopes, enabling real-time estimation of the vehicle's position, velocity, and attitude. However, because INS estimates are obtained through continuous integration of sensor measurements, navigation errors caused by sensor bias and noise accumulate over time. In contrast, the Global Positioning System (GPS) provides absolute position and velocity measurements with high long-term accuracy but at a lower update rate and with susceptibility to signal degradation.

To overcome the limitations of individual sensors, INS and GPS are integrated through sensor fusion techniques. In the integrated framework, the INS provides continuous state propagation, while GPS measurements are used periodically to correct accumulated navigation errors. This complementary relationship enables accurate, stable, and reliable navigation performance during UAV operations. The present chapter describes the sensor measurements, nonlinear dynamic model, and INS/GPS integration architecture that form the foundation of the Adaptive Unscented Kalman Filter (AUKF) based navigation framework developed in the subsequent chapters.

2.1 Navigation Sensor Measurements

The navigation system utilizes measurements obtained from multiple onboard sensors including the Inertial Measurement Unit (IMU), magnetometer, barometric altimeter, and GPS receiver. The IMU provides angular rate and specific force measurements through gyroscopes and accelerometers, respectively. These measurements are used for state propagation within the inertial navigation framework.

Figures below illustrate the sensor measurements collected during the UAV flight experiment.

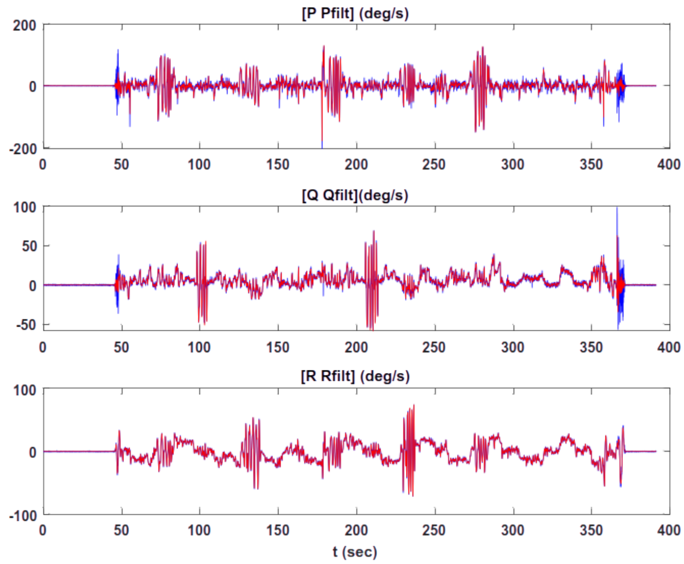

Figure 2.1.1: Measured angular rate components obtained from the gyroscope sensor

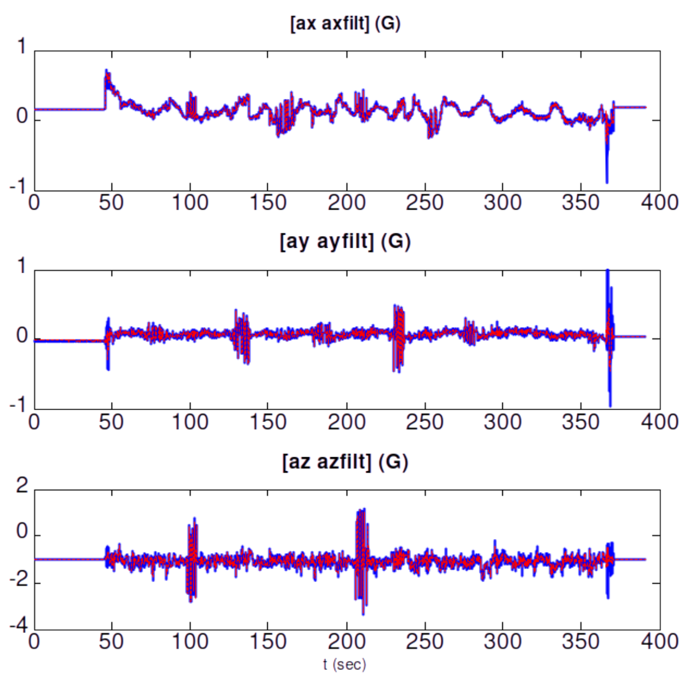

Figure 2.1.2: Measured specific force components obtained from the accelerometer sensor

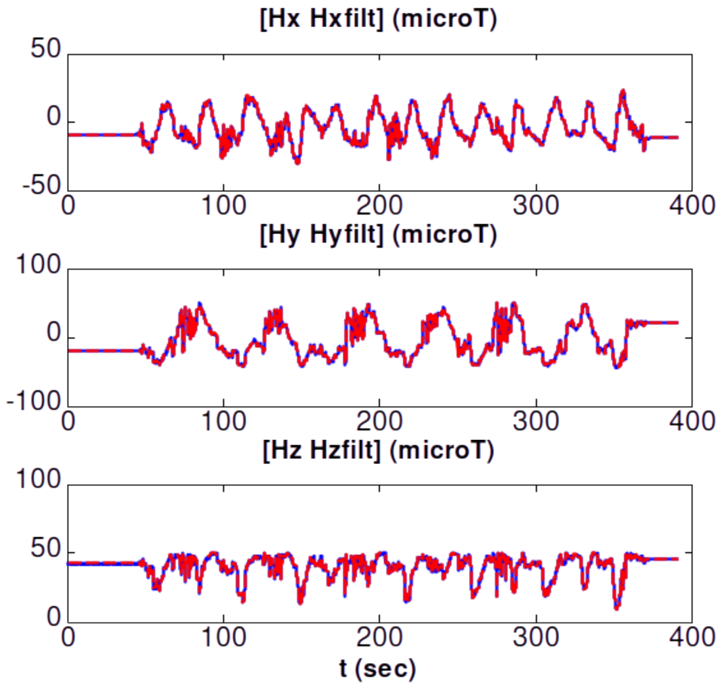

Figure 2.1.3: Magnetometer measurements used for heading estimation

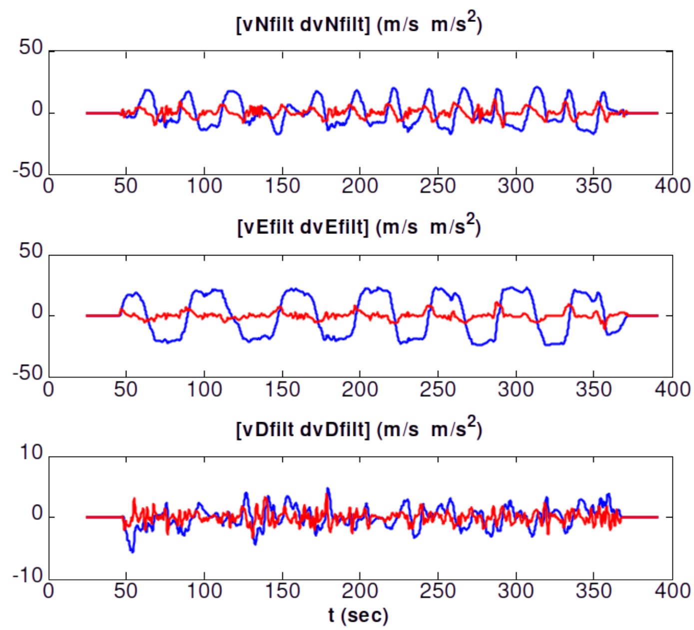

Figure 2.1.4: GPS-derived velocity components in the NED frame

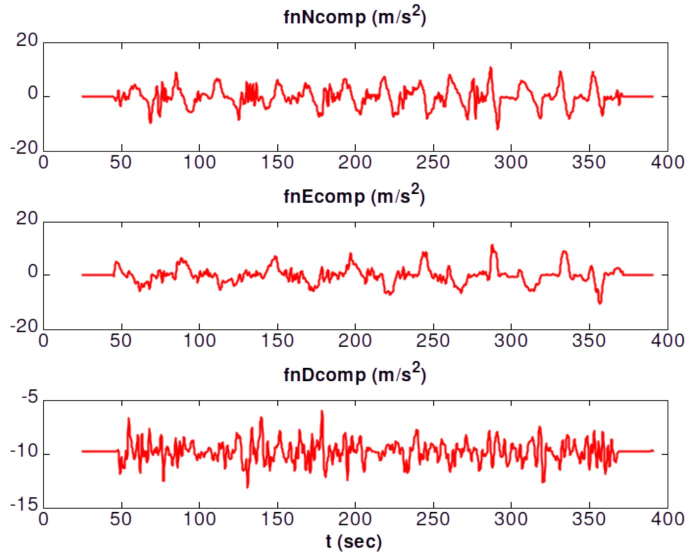

Figure 2.1.5: Functional components in NED navigation frame

The gyroscope measurements provide angular rate information about the UAV body axes, while accelerometer measurements capture the specific forces acting on the vehicle. Magnetometer measurements provide heading reference information and GPS measurements supply position and velocity observations used for navigation correction. These sensor measurements collectively form the inputs to the INS/GPS integrated navigation system.

2.2 System Dynamics

To accurately represent the motion of the UAV, a nonlinear state-space model is developed using position, velocity, attitude, and sensor bias states. The resulting process model serves as the foundation for the navigation filter described in later chapters.

$$x = \begin{bmatrix} \phi & \lambda & h & v_N & v_E & v_D & q_0 & q_1 & q_2 & q_3 \\ b_{a_x} & b_{a_y} & b_{a_z} & b_{g_x} & b_{g_y} & b_{g_z} & & & & \end{bmatrix}^T \tag{2.1}$$

where ϕ, λ, and h denote latitude, longitude, and altitude respectively. The terms v_N, v_E, and v_D represent velocity components in the NED frame. The quaternion states (q_0, q_1, q_2, q_3) represent UAV attitude, while b_a and b_g correspond to accelerometer and gyroscope bias states.

To avoid singularities associated with Euler-angle representation, the UAV attitude is modeled using unit quaternions. The differential equation governing quaternion dynamics is given by

$$\dot{\mathbf{q}} = \frac{1}{2}\begin{bmatrix} 0 & -\boldsymbol{\omega}^T \\ \boldsymbol{\omega} & -[\boldsymbol{\omega}_\times] \end{bmatrix}\begin{bmatrix} s \\ \mathbf{w} \end{bmatrix} \tag{2.2}$$

which can be written in compact form as

$$\dot{\mathbf{q}} = \frac{1}{2}\begin{bmatrix} -\boldsymbol{\omega}^T\mathbf{w} \\ s\boldsymbol{\omega} - \boldsymbol{\omega} \times \mathbf{w} \end{bmatrix} = \frac{1}{2}T(\mathbf{q})\boldsymbol{\omega} \tag{2.3}$$

where $\mathbf{q} = (s, \mathbf{w})$ represents the quaternion vector, $\boldsymbol{\omega} = [P\ Q\ R]^T$ denotes the angular velocity vector, and $[\boldsymbol{\omega}_\times]$ represents the skew-symmetric matrix. Quaternion representation provides smooth and singularity-free attitude estimation during nonlinear UAV maneuvers.

For long-distance navigation, the UAV motion is modeled using nonlinear position and velocity equations in the NED frame. The position kinematics are expressed as

$$\begin{bmatrix} \dot{\phi} \\ \dot{\lambda} \\ \dot{h} \end{bmatrix} = \begin{bmatrix} \frac{1}{M+h} & 0 & 0 \\ 0 & \frac{1}{(N+h)\cos\phi} & 0 \\ 0 & 0 & -1 \end{bmatrix}\begin{bmatrix} v_N \\ v_E \\ v_D \end{bmatrix} \tag{2.4}$$

where M and N denote the meridian and transverse radii of curvature of the Earth respectively.

The nonlinear velocity dynamics are represented as

$$\begin{bmatrix} \dot{v}_N \\ \dot{v}_E \\ \dot{v}_D \end{bmatrix} = \mathbf{f}^n + \mathbf{g}^n + \begin{bmatrix} -v_E\left(2\omega_e + \frac{v_E}{(N+h)\cos\phi}\right)\sin\phi + \frac{v_N v_D}{M+h} \\ v_N \sin\phi\left(2\omega_e + \frac{v_E}{(N+h)\cos\phi}\right) + \frac{v_E v_D}{N+h} \\ -\frac{v_N^2}{M+h} - 2v_E\omega_e\cos\phi - \frac{v_E^2}{N+h} \end{bmatrix} \tag{2.5}$$

where $\mathbf{f}^n = R_b^n(\tilde{\mathbf{f}}^b - \mathbf{b}_a)$ represents the specific force transformed into the navigation frame, $\mathbf{g}^n$ denotes gravity, and ω_e represents the Earth rotation rate.

The quaternion-based attitude dynamics are further expressed as

$$\dot{\mathbf{q}} = \frac{1}{2} \begin{bmatrix} -\mathbf{w}^T \\ sI_3 + [\mathbf{w}_\times] \end{bmatrix} \left(\tilde{\boldsymbol{\omega}}^b - \mathbf{b}_\omega - R_n^b(\boldsymbol{\omega}_{ie}^n + \boldsymbol{\omega}_{en}^n) \right) \tag{2.6}$$

where $\tilde{\boldsymbol{\omega}}^b$ denotes measured angular velocity, $\mathbf{b}_\omega$ represents gyroscope bias, and $(\boldsymbol{\omega}_{ie}^n + \boldsymbol{\omega}_{en}^n)$ correspond to Earth rotation and transport rate components.

The accelerometer and gyroscope bias dynamics are modeled as slowly varying random processes and are represented as

$$\dot{\mathbf{b}}_a = 0, \qquad \dot{\mathbf{b}}_\omega = 0 \tag{2.7}$$

These equations collectively define the nonlinear INS process model used for UAV state propagation.

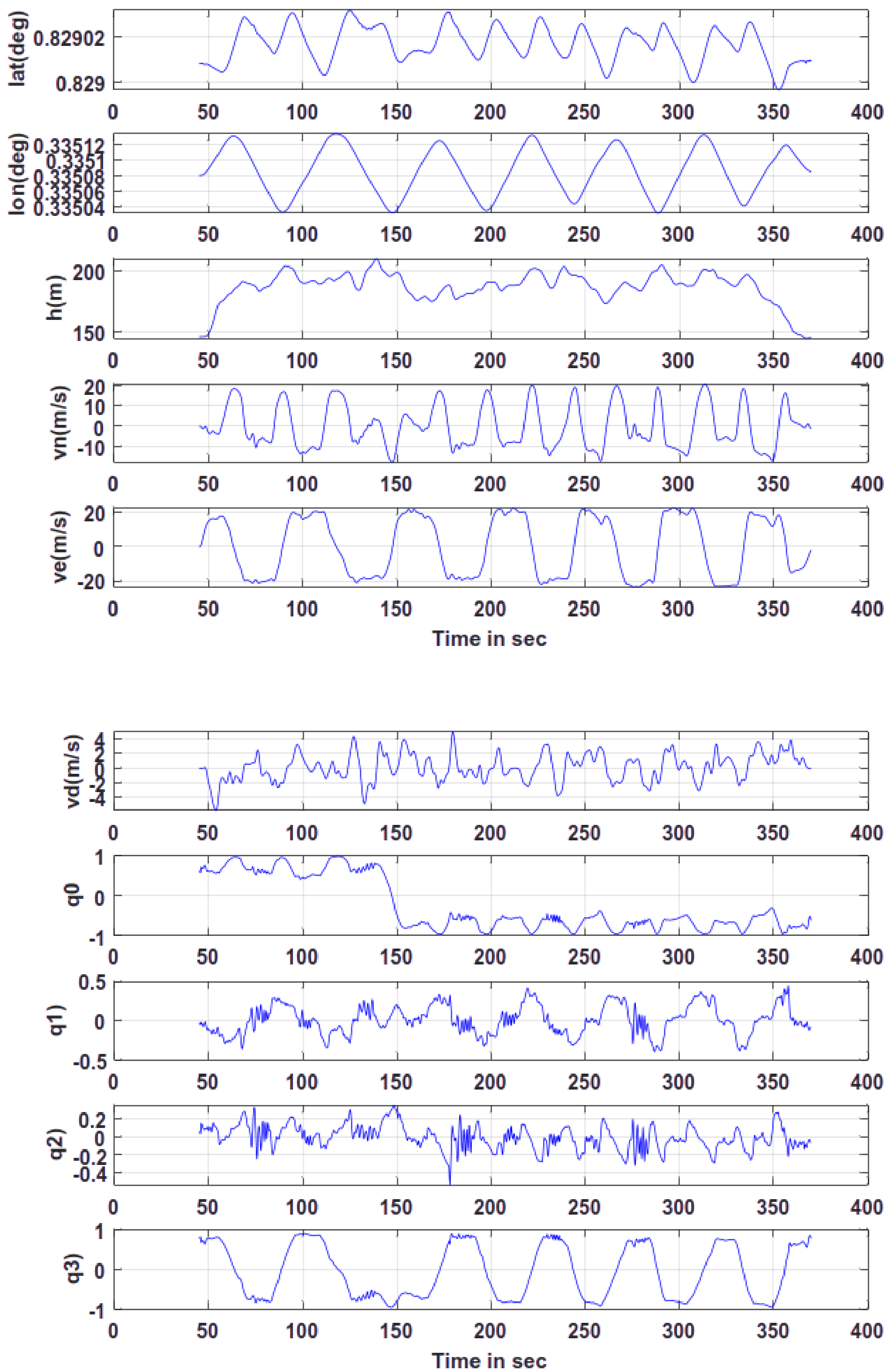

Figure 2.2.1: Estimated navigation states including position, velocity, and quaternion attitude components obtained from INS/GPS integration

2.3 INS/GPS Sensor Fusion Framework

The INS/GPS sensor fusion framework combines the prediction capability of the INS with the correction capability of GPS measurements to achieve robust navigation performance. The INS continuously propagates the UAV states at a high sampling rate using IMU measurements. However, due to continuous integration of noisy sensor data, INS-only navigation experiences drift accumulation over time. GPS measurements are therefore incorporated periodically to correct these errors and maintain long-term navigation accuracy.

The coupled GPS/INS kinematic model is represented using the following position update equation

$$\dot{\mathbf{p}} = \begin{bmatrix} \frac{1}{M+h} & 0 & 0 \\ 0 & \frac{1}{(N+h)\cos\phi} & 0 \\ 0 & 0 & -1 \end{bmatrix} \mathbf{v} \tag{2.8}$$

where $\mathbf{p}$ and $\mathbf{v}$ denote the position and velocity vectors respectively.

The velocity update equation is expressed as

$$\dot{\mathbf{v}} = \mathbf{f}^n + \mathbf{g}^n + \text{nonlinear coupling terms} \tag{2.9}$$

where the nonlinear coupling terms account for Coriolis effects, Earth rotation, and transport rate dynamics.

The angular velocity transformation used in the integration process is represented as

$$\boldsymbol{\omega} = \tilde{\boldsymbol{\omega}}^b - \mathbf{b}_\omega - R_b^n(\boldsymbol{\omega}_{ie}^n + \boldsymbol{\omega}_{en}^n) \tag{2.10}$$

with

$$\boldsymbol{\omega}_{ie}^n + \boldsymbol{\omega}_{en}^n = \begin{bmatrix} \cos\phi \\ 0 \\ -\sin\phi \end{bmatrix} \omega_e + \begin{bmatrix} \frac{v_E}{N+h} \\ -\frac{v_N}{M+h} \\ -\frac{v_E \tan\phi}{N+h} \end{bmatrix} \tag{2.11}$$

The sensor fusion process is implemented using the Unscented Kalman Filter

(UKF), which is suitable for nonlinear UAV dynamics. During the prediction stage, the UKF propagates sigma points through the nonlinear INS process model using IMU measurements. In the correction stage, GPS observations are compared with predicted states to generate innovation errors, which are then used to update the state estimates and covariance matrix recursively.

Through continuous interaction between INS prediction and GPS correction, the integrated navigation framework improves estimation accuracy, reduces sensor uncertainty, and provides stable UAV navigation under varying operating conditions. This integrated estimation architecture forms the basis for the subsequent GPS-denied navigation framework discussed in the following chapters.

2.4 Chapter Summary

This chapter presented the INS/GPS integrated navigation framework employed for UAV state estimation. The sensor measurements used by the navigation system were first introduced, followed by the nonlinear dynamic model describing position, velocity, attitude, and sensor bias states. The INS/GPS sensor fusion architecture was then discussed, demonstrating how inertial propagation and GPS correction are combined to provide accurate navigation performance.

CHAPTER 3

Adaptive Unscented Kalman Filter

The UAV navigation system considered in this work exhibits highly nonlinear dynamics due to coupled translational and rotational motion, sensor noise, and uncertainty in inertial measurements. Conventional linear estimation techniques are unable to accurately handle such nonlinear behavior, especially during aggressive maneuvers and GPS-denied conditions. To overcome these limitations, the Adaptive Unscented Kalman Filter (AUKF) is employed for nonlinear state estimation and sensor fusion.

The Unscented Kalman Filter (UKF) is based on the Unscented Transform, which propagates a set of deterministically selected sigma points through the nonlinear system dynamics instead of linearizing the model using Jacobian matrices. The adaptive version of the UKF further enhances estimation performance by allowing the process and measurement noise covariance matrices to be updated online based on innovation statistics. This capability is particularly useful for UAV navigation applications where sensor characteristics and environmental conditions may vary significantly during flight. The AUKF therefore provides a robust and adaptive framework for accurate nonlinear state estimation under both nominal and GPS-denied operating conditions.

3.1 State Space Representation

The nonlinear UAV system is represented in discrete-time state-space form as

$$x_k = f(x_{k-1}, u_k) + w_k \tag{3.1}$$

$$z_k = h(x_k) + v_k \tag{3.2}$$

where x_k represents the system state vector, u_k denotes the control input, and z_k represents the measurement vector. The nonlinear functions $f(\cdot)$ and $h(\cdot)$ correspond to the process and measurement models respectively. The terms w_k and v_k denote process and measurement noise with covariance matrices Q_k and R_k.

The state vector used in the navigation framework is expressed as

$$x = \begin{bmatrix} \phi & \lambda & h & v_N & v_E & v_D & q_0 & q_1 & q_2 & q_3 \\ b_{ax} & b_{ay} & b_{az} & b_{gx} & b_{gy} & b_{gz} & & & & \end{bmatrix}^T \tag{3.3}$$

which includes position, velocity, attitude quaternion, accelerometer bias, and gyroscope bias states.

3.2 Unscented Transform and Sigma Point Generation

The UKF begins by generating sigma points around the current state estimate. For an n-dimensional state vector, a total of $2n + 1$ sigma points are generated using the covariance matrix.

The scaling parameter is computed as

$$\lambda = \alpha^2(n + \kappa) - n \tag{3.4}$$

where α, β, and κ are tuning parameters of the Unscented Transform.

The sigma points are generated as

$$\chi_0 = \hat{x}_{k-1} \tag{3.5}$$

$$\chi_i = \hat{x}_{k-1} + \left(\sqrt{(n + \lambda)P_{k-1}}\right)_i \tag{3.6}$$

$$\chi_{i+n} = \hat{x}_{k-1} - \left(\sqrt{(n + \lambda)P_{k-1}}\right)_i \tag{3.7}$$

for $i = 1, 2, \ldots, n$, where P_{k-1} is the state covariance matrix.

The associated sigma point weights are defined as

$$W_0^{(m)} = \frac{\lambda}{n + \lambda} \tag{3.8}$$

$$W_0^{(c)} = \frac{\lambda}{n + \lambda} + (1 - \alpha^2 + \beta) \tag{3.9}$$

$$W_i^{(m)} = W_i^{(c)} = \frac{1}{2(n+\lambda)} \tag{3.10}$$

for $i = 1, 2, \ldots, 2n$.

3.3 AUKF Filtering Procedure

The Adaptive Unscented Kalman Filter (AUKF) operates recursively through prediction and measurement update stages. During the prediction stage, the sigma points are propagated through the nonlinear process model to obtain the predicted state estimate and covariance. Subsequently, measurement information is incorporated during the update stage to correct the predicted states and reduce estimation uncertainty. This recursive estimation framework enables accurate tracking of UAV states under nonlinear operating conditions.

3.3.1 Prediction Stage

In the prediction stage, each sigma point is propagated through the nonlinear process model using the UAV dynamic equations.

$$\chi_{k|k-1}^{(i)} = f(\chi_{k-1}^{(i)}, u_k) \tag{3.11}$$

The predicted state mean is obtained as

$$\hat{x}_{k|k-1} = \sum_{i=0}^{2n} W_i^{(m)} \chi_{k|k-1}^{(i)} \tag{3.12}$$

The predicted covariance matrix is computed as

$$P_{k|k-1} = \sum_{i=0}^{2n} W_i^{(c)} \left(\chi_{k|k-1}^{(i)} - \hat{x}_{k|k-1}\right) \left(\chi_{k|k-1}^{(i)} - \hat{x}_{k|k-1}\right)^T + Q_k \tag{3.13}$$

where Q_k represents the process noise covariance matrix.

3.3.2 Measurement Update Stage

After prediction, the sigma points are transformed through the nonlinear measurement model.

$$Z_k^{(i)} = h(\chi_{k|k-1}^{(i)}) \tag{3.14}$$

The predicted measurement mean is computed as

$$\hat{z}_k = \sum_{i=0}^{2n} W_i^{(m)} Z_k^{(i)} \tag{3.15}$$

The innovation covariance matrix is expressed as

$$S_k = \sum_{i=0}^{2n} W_i^{(c)} (Z_k^{(i)} - \hat{z}_k)(Z_k^{(i)} - \hat{z}_k)^T + R_k \tag{3.16}$$

where R_k denotes the measurement noise covariance matrix.

The cross-covariance matrix between the states and measurements is given by

$$P_{xz} = \sum_{i=0}^{2n} W_i^{(c)} (\chi_{k|k-1}^{(i)} - \hat{x}_{k|k-1})(Z_k^{(i)} - \hat{z}_k)^T \tag{3.17}$$

The Kalman gain is then computed as

$$K_k = P_{xz} S_k^{-1} \tag{3.18}$$

Using the innovation between actual and predicted measurements, the state estimate is updated as

$$\hat{x}_k = \hat{x}_{k|k-1} + K_k(z_k - \hat{z}_k) \tag{3.19}$$

The updated covariance matrix is given by

$$P_k = P_{k|k-1} - K_k S_k K_k^T \tag{3.20}$$

3.4 Adaptive Covariance Estimation

In practical UAV navigation systems, process and measurement noise characteristics may vary with time due to environmental disturbances, sensor degradation, and GPS outages. Therefore, adaptive covariance estimation is incorporated into the UKF framework to improve robustness and estimation accuracy.

The process and measurement covariance matrices are updated adaptively using innovation statistics as

$$Q_k = \gamma Q_{k-1} + (1 - \gamma)\Delta Q_k \tag{3.21}$$

$$R_k = \gamma R_{k-1} + (1 - \gamma)\Delta R_k \tag{3.22}$$

where γ represents the forgetting factor and ΔQ_k, ΔR_k denote covariance correction terms obtained from innovation residuals.

The adaptive mechanism enables the filter to respond dynamically to changing system uncertainties and improves estimation performance under nonlinear and uncertain operating conditions.

3.5 Filter Performance Results

The effectiveness of the proposed AUKF is evaluated using UAV flight data. Figures 3.5.1–3.5.4 present the estimation performance of the AUKF in terms of sensor bias estimation and attitude tracking.

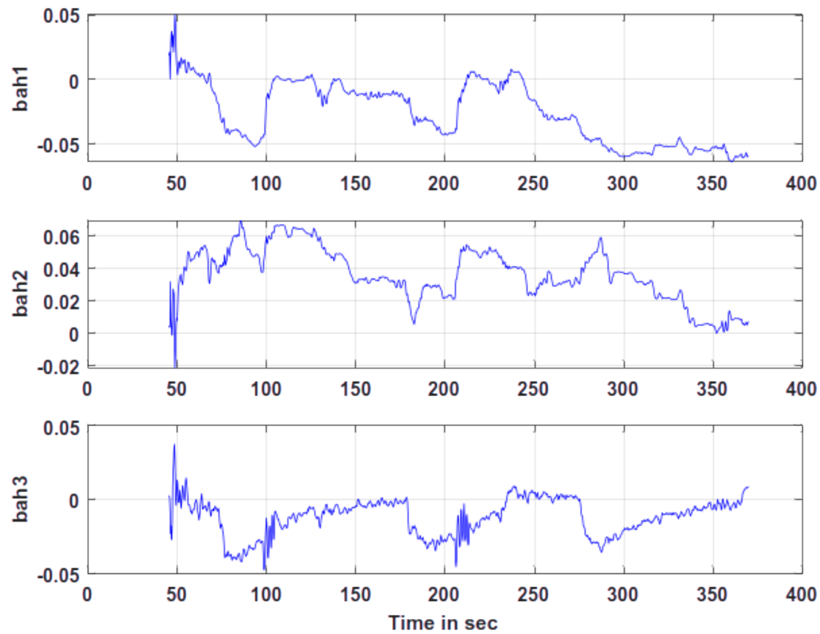

Figure 3.5.1: Accelerometer Bias Estimation

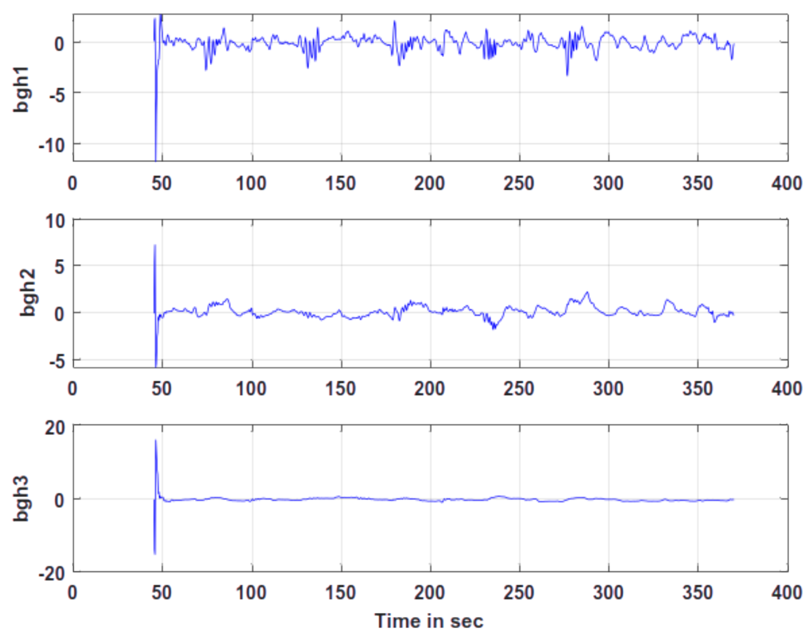

Figure 3.5.2: Gyroscope Bias Estimation

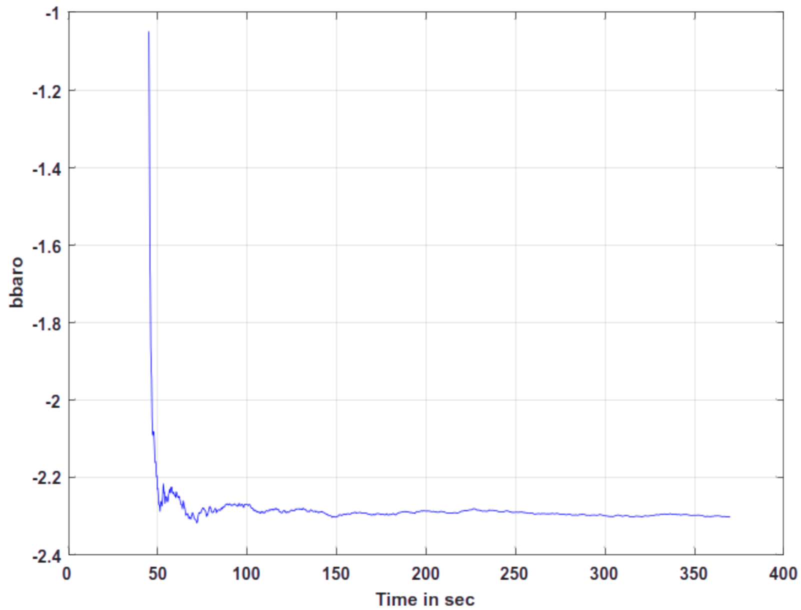

Figure 3.5.3: Barometer Bias Estimation

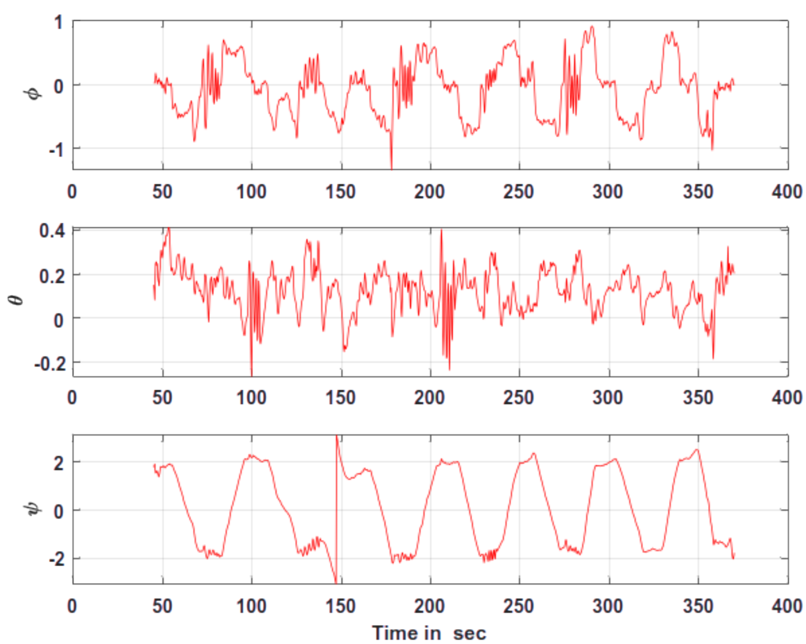

Figure 3.5.4: Estimated Attitude Response

3.6 Chapter Summary

This chapter presented the Adaptive Unscented Kalman Filter (AUKF) employed for nonlinear UAV state estimation. The complete filtering procedure including sigma point generation, prediction, measurement update, and adaptive covariance estimation was discussed in detail. By propagating sigma points through nonlinear system dynamics without Jacobian-based linearization, the AUKF improves estimation accuracy and robustness under nonlinear operating conditions. The adaptive covariance tuning mechanism further enhances filter performance in the presence of uncertain sensor noise and disturbances, making the AUKF suitable for reliable UAV navigation in GPS-denied environments.

CHAPTER 4

Neural Network-Based Error Compensation

The performance of INS/GPS integrated navigation systems degrades significantly during GPS outages due to the accumulation of inertial sensor errors. Although nonlinear filtering algorithms such as the Adaptive Unscented Kalman Filter (AUKF) improve estimation accuracy, long-duration GPS-denied operation still leads to drift in position, velocity, and attitude estimates. To overcome this limitation, a Neural Network (NN) is incorporated into the navigation framework to learn the nonlinear error characteristics of the INS and generate correction information during GPS-denied operation.

The Neural Network acts as a data-driven estimation model that learns the relationship between inertial sensor measurements and navigation drift errors. During normal GPS availability, the INS/GPS integrated system generates corrected navigation states, which are used to train the Neural Network. Once trained, the NN predicts navigation errors during GPS outages and generates pseudo-GPS measurements for maintaining navigation continuity. This hybrid approach combines the stability of model-based filtering with the adaptability of machine learning for robust UAV navigation.

In the proposed navigation framework, the Neural Network serves as an auxiliary estimator that complements the Adaptive Unscented Kalman Filter during GPS-denied operation. By learning navigation error patterns from historical INS/GPS data, the network is capable of predicting future drift characteristics and generating correction information in real time. This capability enables the navigation system to maintain acceptable estimation accuracy even when external GPS measurements become unavailable.

4.1 Neural Network Architecture

The Neural Network used in this work is a feedforward multilayer perceptron (MLP) designed to estimate navigation errors under nonlinear operating conditions. The network consists of an input layer, hidden layers, and an output layer. The input layer receives IMU measurements and estimated navigation states, while the output layer predicts the corresponding INS drift errors.

The input vector to the Neural Network is expressed as

$$X_{in} = \begin{bmatrix} a_x & a_y & a_z & p & q & r & v_N & v_E & v_D & q_0 \\ q_1 & q_2 & q_3 \end{bmatrix}^T \tag{4.1}$$

where a_x, a_y, and a_z represent accelerometer measurements, (p, q, r) denote body angular rates obtained from gyroscopes, (v_N, v_E, v_D) represent velocity states in the NED frame, and (q_0, q_1, q_2, q_3) correspond to quaternion attitude states.

The output vector of the Neural Network is represented as

$$Y_{out} = \begin{bmatrix} \Delta\phi & \Delta\lambda & \Delta h & \Delta v_N & \Delta v_E & \Delta v_D \end{bmatrix}^T \tag{4.2}$$

where $\Delta\phi$, $\Delta\lambda$, and Δh denote position estimation errors, while Δv_N, Δv_E, and Δv_D represent velocity correction terms.

The hidden layers utilize nonlinear activation functions to capture the complex nonlinear relationship between IMU measurements and navigation drift. The neuron output is expressed as

$$y_j = f\left(\sum_{i=1}^{n} w_{ij} x_i + b_j\right) \tag{4.3}$$

where x_i denotes the input vector, w_{ij} represents the connection weights, b_j is the bias term, and $f(\cdot)$ denotes the activation function.

The output layer uses a linear activation function since the network performs continuous-valued regression for navigation error prediction.

4.2 Training Process

The Neural Network is trained using supervised learning based on data collected during normal INS/GPS operation. During GPS availability, the INS/GPS integrated system provides corrected navigation states, while the INS continuously propagates navigation states using IMU measurements. The difference between INS-only estimates and corrected navigation states represents the navigation drift error.

The target error vector used for training is defined as

$$E_k = X_{INS,k} - X_{GPS,k} \tag{4.4}$$

where $X_{INS,k}$ represents the INS-estimated state vector and $X_{GPS,k}$ denotes the corresponding GPS-corrected navigation states.

The Neural Network learns the nonlinear relationship between IMU measurements and INS drift using these error values as target outputs. The network parameters are optimized by minimizing the Mean Squared Error (MSE) loss function given by

$$J = \frac{1}{N}\sum_{k=1}^{N}(E_k - \hat{E}_k)^2 \tag{4.5}$$

where E_k denotes the actual navigation error and $\hat{E}_k$ represents the Neural Network predicted error.

The backpropagation algorithm combined with gradient descent optimization is used to update the network weights iteratively. The weight update equation is expressed as

$$w_{ij}(k+1) = w_{ij}(k) - \eta\frac{\partial J}{\partial w_{ij}} \tag{4.6}$$

where η represents the learning rate.

Training is continued until the loss function converges and the Neural Network achieves satisfactory prediction accuracy for navigation error estimation.

The trained Neural Network was evaluated using the navigation error data obtained from the INS/GPS integrated navigation system. The objective was to assess the ability of the network to learn the nonlinear relationship between inertial sensor measurements and navigation drift. Figures 4.2.1–4.2.4 illustrate the actual position and velocity errors extracted from the navigation dataset, while Figures 4.2.5–4.2.8 show the corresponding Neural Network predictions. The close agreement between the actual and predicted errors demonstrates the capability of the Neural Network to accurately model INS error characteristics and provide reliable correction information during GPS-denied operation.

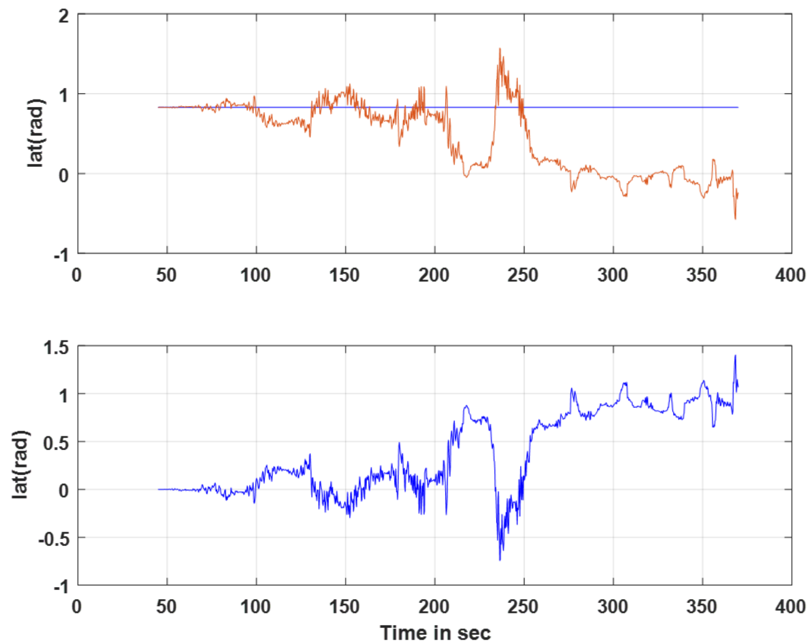

Figure 4.2.1: Latitude estimation error obtained from INS/GPS navigation

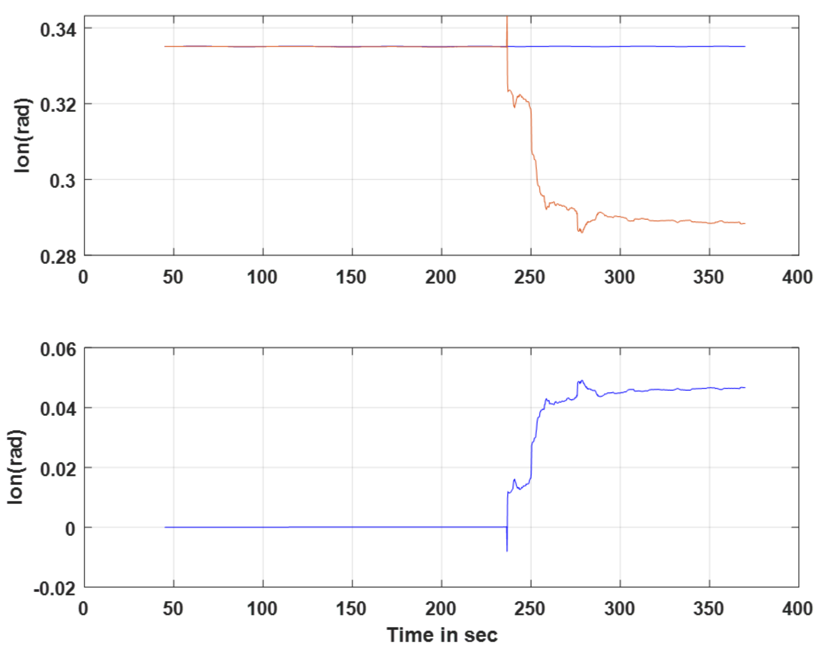

Figure 4.2.2: Longitude estimation error obtained from INS/GPS navigation

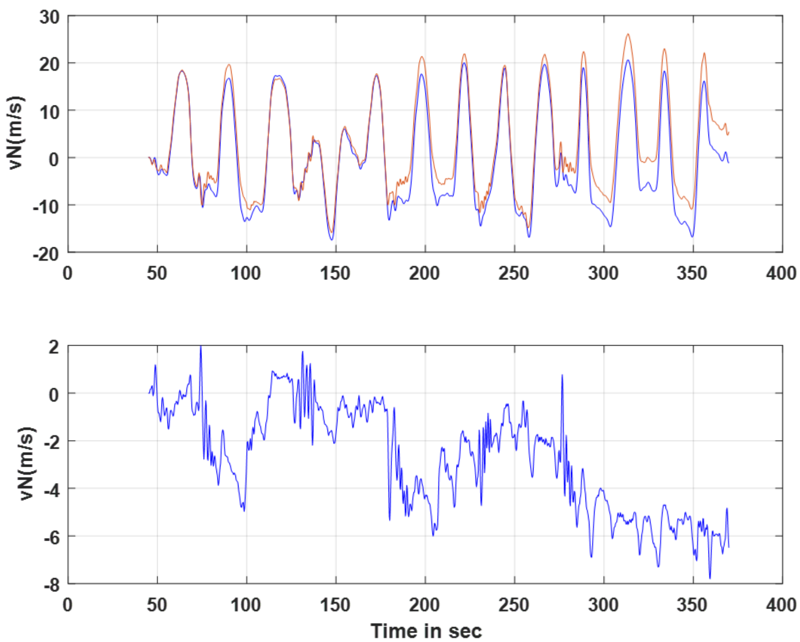

Figure 4.2.3: North velocity estimation error obtained from INS/GPS navigation

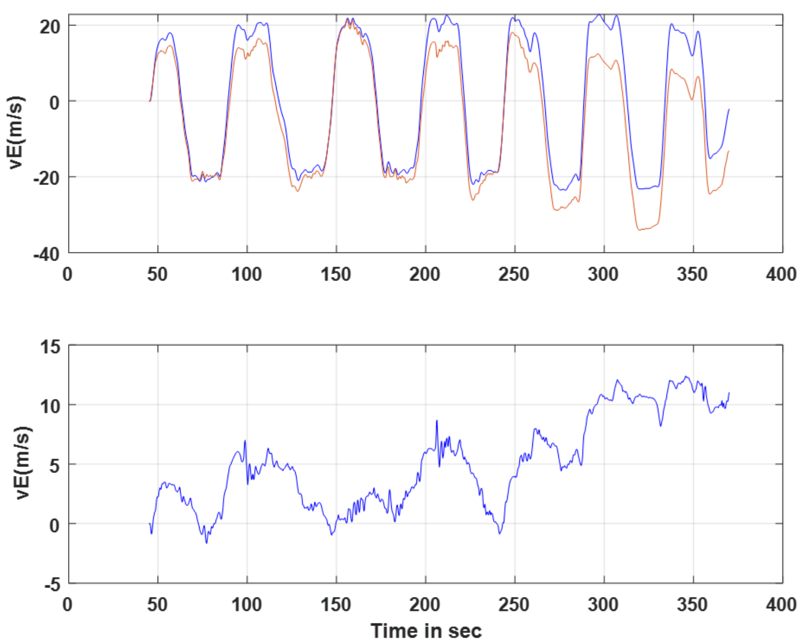

Figure 4.2.4: East velocity estimation error obtained from INS/GPS navigation

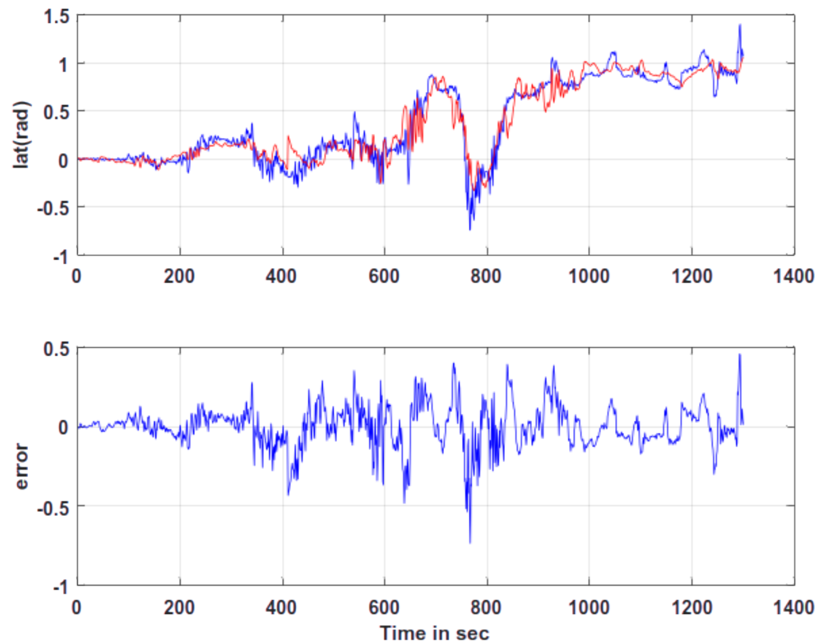

Figure 4.2.5: Neural Network prediction of latitude error

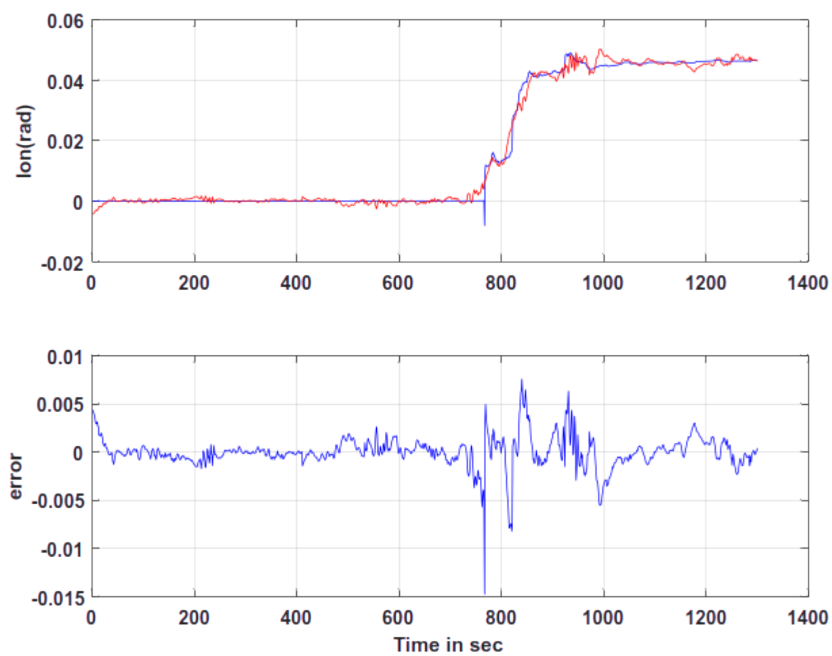

Figure 4.2.6: Neural Network prediction of longitude error

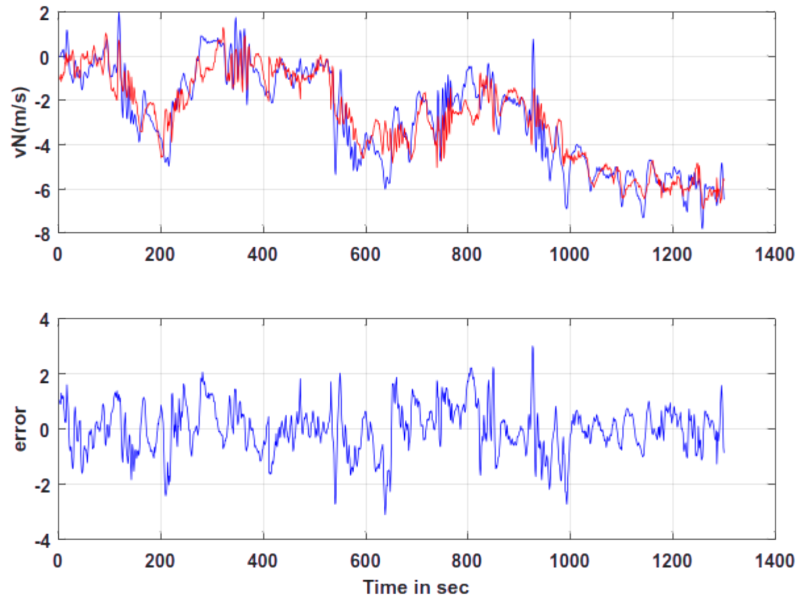

Figure 4.2.7: Neural Network prediction of north velocity error

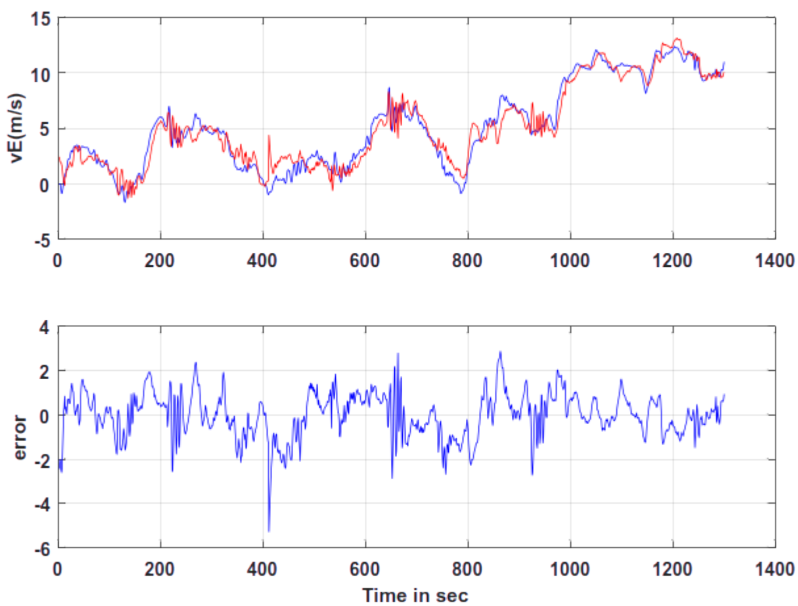

Figure 4.2.8: Neural Network prediction of down velocity error

4.3 Generation of Pseudo-GPS Measurements

During normal operating conditions, both INS and GPS measurements are available for navigation. The INS continuously propagates the UAV states using IMU measurements, while GPS provides accurate position and velocity updates. Since INS-based estimation accumulates drift over time, the difference between the INS-predicted states and GPS measurements represents the navigation error associated with inertial sensor drift.

These navigation errors are extracted and used as target outputs for Neural Network training. The Neural Network learns the nonlinear relationship between inertial sensor measurements and the corresponding INS drift characteristics using supervised learning.

After the training process is completed, the Neural Network is utilized during GPS-denied conditions. In the absence of GPS measurements, the trained network predicts navigation errors using the current IMU measurements and estimated navigation states as inputs. These predicted errors are then used to generate pseudo-GPS measurements that approximate the unavailable GPS updates.

$$Z_{pseudo} = X_{INS} - \hat{E}_{NN} \tag{4.7}$$

where $\hat{E}_{NN}$ represents the navigation error predicted by the Neural Network.

The generated pseudo-GPS measurements were compared with the navigation states obtained during normal GPS operation to evaluate their effectiveness. Figures 4.3.1 and 4.3.2 present the generated pseudo-GPS position and velocity measurements. The results indicate that the Neural Network successfully reconstructs the missing GPS information, thereby providing useful correction data to the navigation filter during GPS outages.

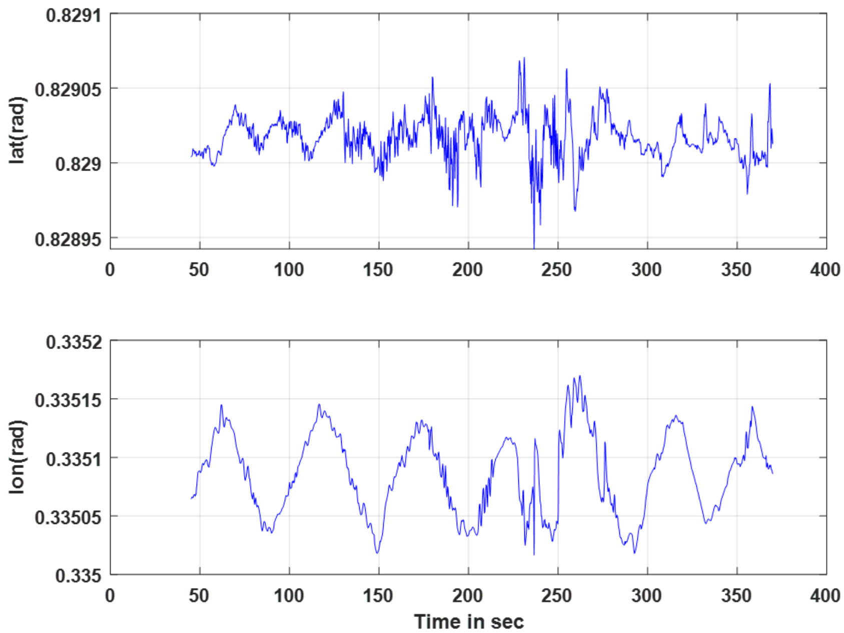

Figure 4.3.1: Generated pseudo-GPS position measurements

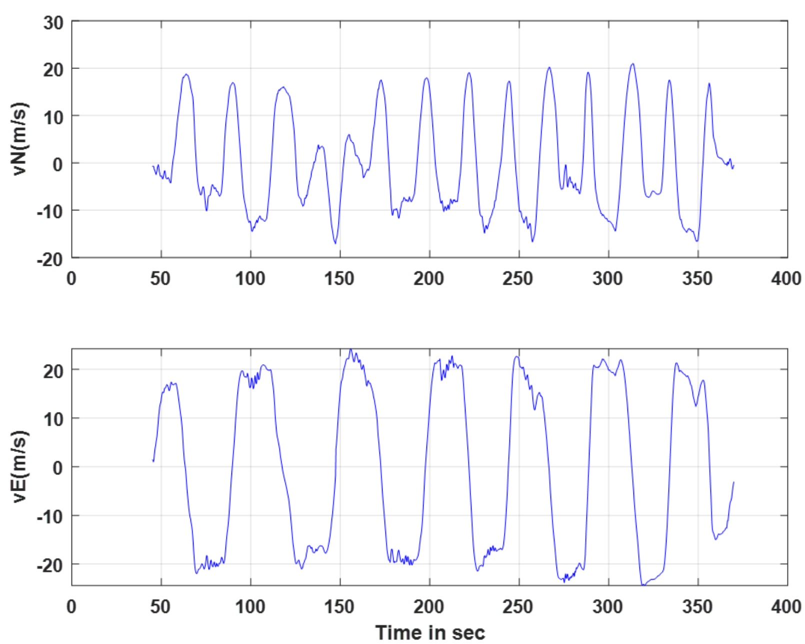

Figure 4.3.2: Generated pseudo-GPS velocity measurements

4.4 Chapter Summary

This chapter presented the Neural Network-based error compensation framework developed for GPS-denied UAV navigation. The Neural Network architecture, training methodology, and navigation error modeling process were discussed in detail. The network was trained using INS/GPS navigation errors and subsequently employed to predict INS drift during GPS outages. The predicted errors were used to generate pseudo-GPS measurements, enabling continuous navigation correction in the absence of GPS observations. The integration of machine learning with nonlinear filtering improves navigation continuity and reduces long-term estimation errors during GPS-denied operation.

CHAPTER 5

Experimental Results and Discussion

This chapter presents the state estimation performance of the proposed UAV navigation framework under different operating conditions. The objective is to evaluate the effectiveness of the INS/GPS integrated navigation system and analyze the impact of GPS outages on estimation accuracy. Three different navigation scenarios are considered for performance evaluation.

In the first case, the UAV operates under normal GPS availability where the Adaptive Unscented Kalman Filter (AUKF) utilizes both INS and GPS measurements for state estimation. In the second case, GPS measurements are completely unavailable and the UAV relies only on standalone INS propagation, resulting in drift accumulation. In the third case, the proposed Neural Network-generated pseudo-GPS measurements are incorporated during GPS outages to compensate for INS drift and improve navigation accuracy.

The performance of the navigation framework is analyzed using position estimation, velocity estimation, attitude response, trajectory tracking, and estimation error characteristics under each operating condition.

5.1 Case 1: INS/GPS Integrated Navigation

In the first scenario, the UAV navigation system operates under normal GPS availability. The Adaptive Unscented Kalman Filter integrates high-rate INS measurements with low-rate GPS observations to provide accurate and stable state estimation.

The INS continuously propagates the UAV states using accelerometer and gyroscope measurements, while GPS measurements periodically correct accumulated navigation errors. Through recursive prediction and correction stages, the AUKF provides accurate estimation of position, velocity, attitude, and sensor bias states.

Under GPS-assisted operation, the navigation errors remain bounded due to periodic correction updates from GPS observations. The integrated INS/GPS framework therefore provides accurate trajectory tracking and stable navigation performance.

The estimated position, velocity, and attitude responses obtained using the INS/GPS integrated navigation framework are shown in the following figures. The plots illustrate the effectiveness of the AUKF in tracking the UAV states accurately under normal GPS availability.

The estimated states obtained under GPS availability serve as the reference navigation solution for subsequent GPS-denied analysis and Neural Network training.

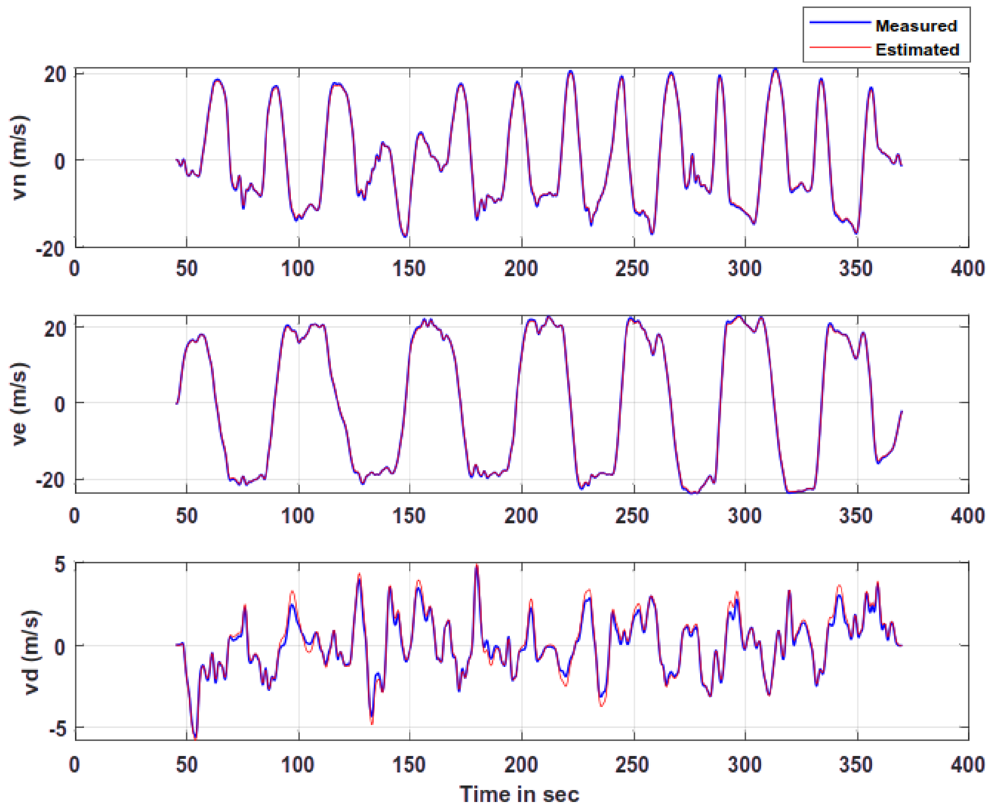

Figure 5.1.1: Comparison of measured and estimated velocity trajectories

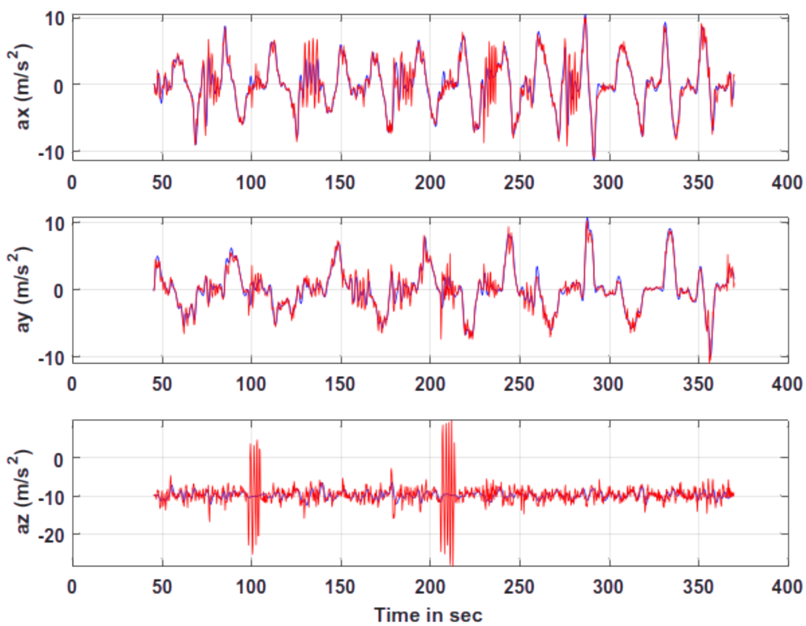

Figure 5.1.2: Comparison of measured and estimated acceleration components

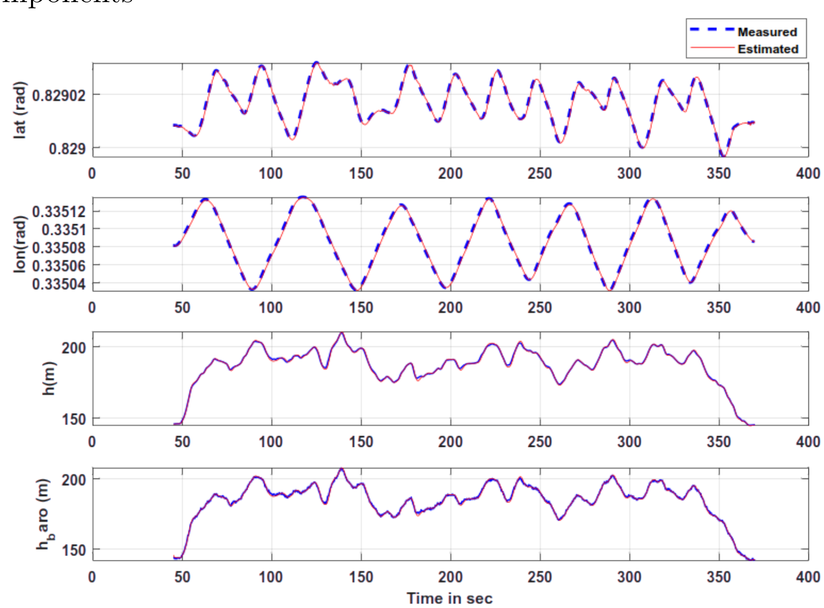

Figure 5.1.3: Comparison of measured and estimated position trajectories

The results demonstrate that the AUKF accurately tracks the UAV motion during normal GPS availability. The estimated position and velocity states closely follow the measured GPS trajectories, while the estimated accelerations remain consistent with the sensor observations. These results confirm the effectiveness of the nonlinear filtering framework for integrated INS/GPS navigation under nominal operating conditions.

5.2 Case 2: Standalone INS Navigation During GPS Outage

In the second scenario, GPS measurements are completely unavailable and the UAV navigation system relies solely on INS propagation. During GPS outages, the filter prediction stage continues using IMU measurements; however, no external correction measurements are available for updating the estimated states.

Since the INS performs continuous integration of accelerometer and gyroscope measurements, even small sensor biases and noise accumulate over time. This accumulation results in increasing drift in position, velocity, and attitude estimates.

As the GPS outage duration increases, the estimation errors become significantly larger, resulting in degradation of navigation accuracy and system reliability. The standalone INS case therefore highlights the limitations of inertial-only navigation during long-duration GPS-denied operation.

The standalone INS estimation results during GPS outage are shown in the following figures. Due to continuous integration of inertial sensor measurements without external correction updates, the estimation errors increase progressively with time, resulting in noticeable drift in the navigation states.

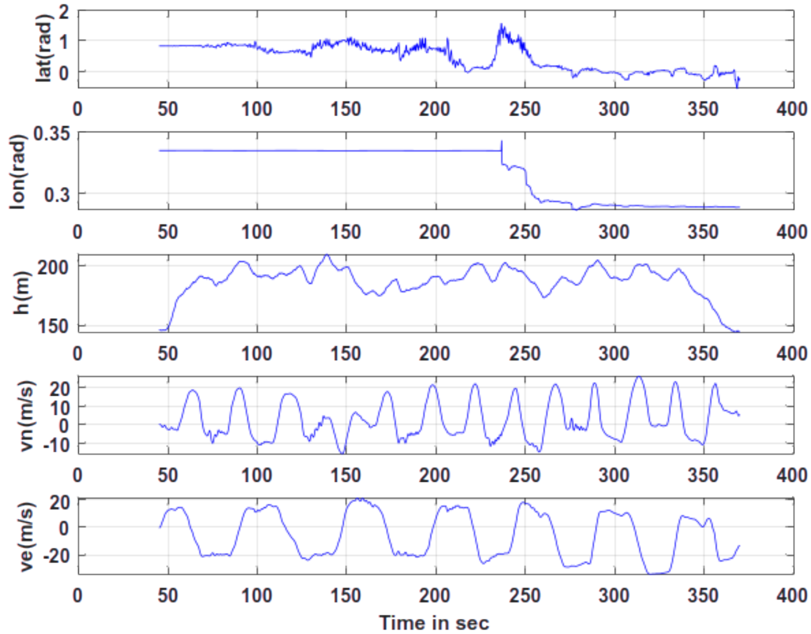

Figure 5.2.1: State Estimation of lat, lon, h, vN and vE without GPS

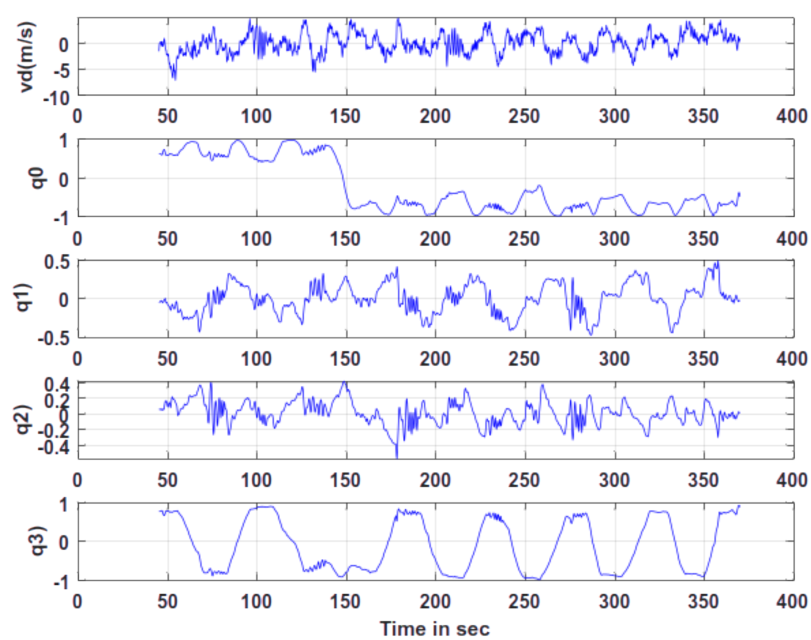

Figure 5.2.2: State Estimation of vD, q0, q1, q2 and q3 without GPS

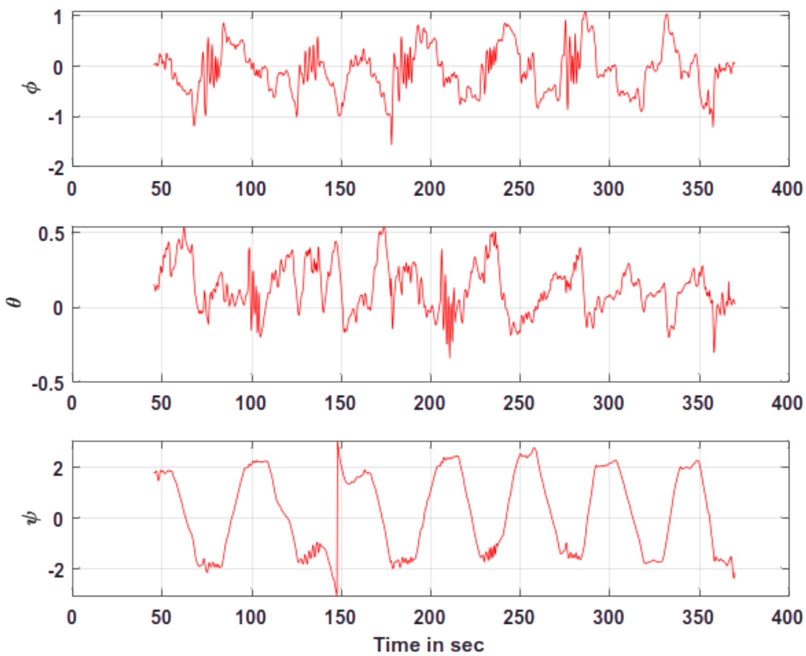

Figure 5.2.3: Attitudes Estimated without GPS Data

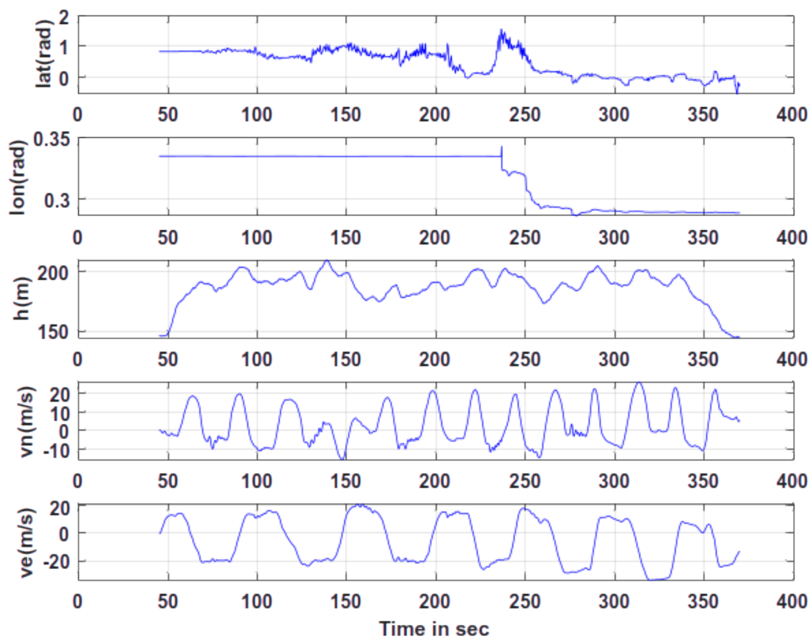

Figure 5.2.4: Measured and Estimated h, vD

The results clearly demonstrate the limitations of standalone INS navigation during GPS-denied operation. In the absence of external correction measurements, sensor biases and noise accumulate through continuous integration, leading to increasing position and velocity errors. Consequently, the navigation solution gradually diverges from the true trajectory, highlighting the need for additional correction mechanisms during extended GPS outages.

5.3 Case 3: Navigation Using Pseudo-GPS Measurements

In the third scenario, the proposed Neural Network-based pseudo-GPS generation framework is utilized during GPS outages. Instead of relying solely on INS propagation, the trained Neural Network predicts navigation drift errors using IMU measurements and estimated navigation states.

The predicted errors are used to generate pseudo-GPS measurements that approximate the unavailable GPS updates. These virtual corrections help reduce INS drift and improve navigation continuity during GPS-denied operation.

By continuously generating pseudo-GPS measurements, the proposed framework maintains bounded estimation errors and improves trajectory tracking performance compared with standalone INS navigation. The Neural Network-generated corrections significantly reduce long-term drift accumulation and enhance navigation robustness under uncertain operating conditions.

The state estimation results obtained using the proposed pseudo-GPS-assisted framework are shown in the following figures. The Neural Network-generated pseudo-measurements help reduce INS drift and improve navigation continuity during GPS-denied operation.

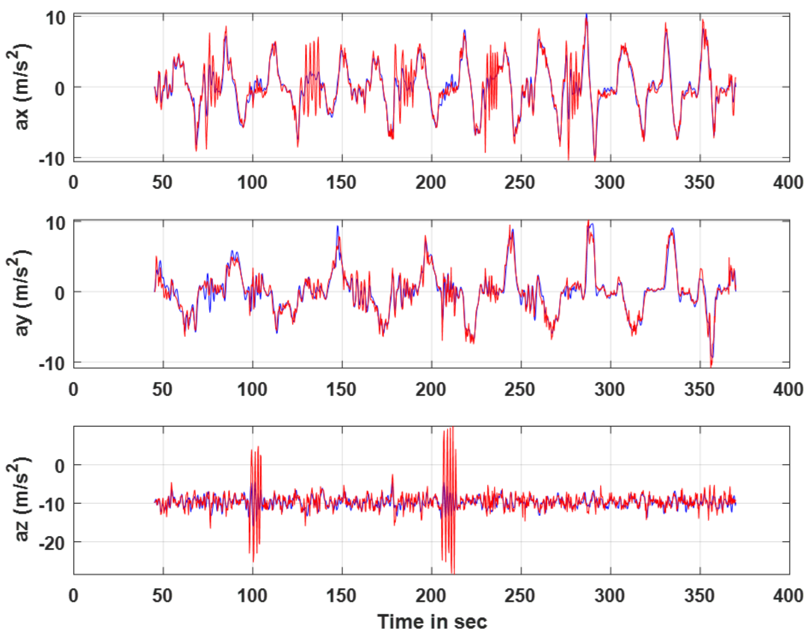

Figure 5.3.1: Comparison of measured and estimated acceleration components

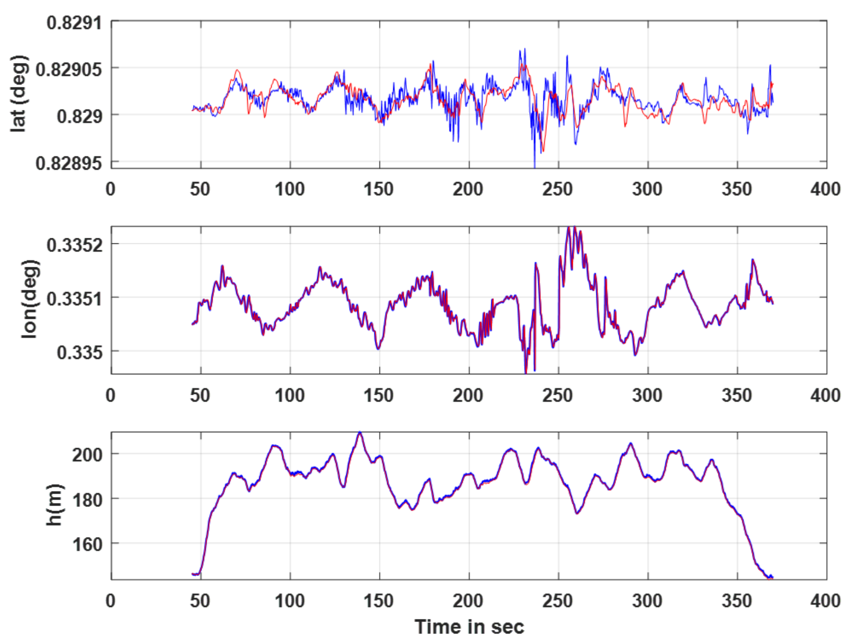

Figure 5.3.2: Comparison of measured and estimated position trajectories

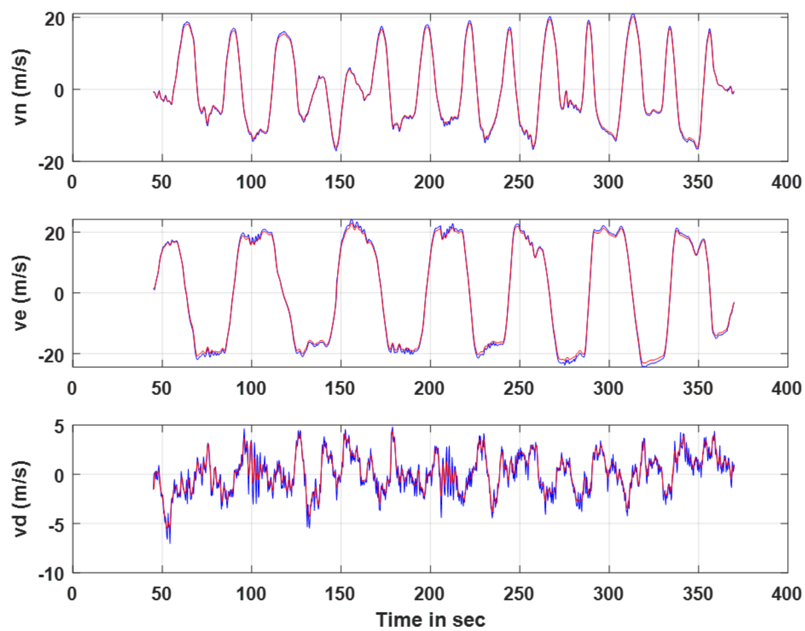

Figure 5.3.3: Comparison of measured and estimated velocity trajectories

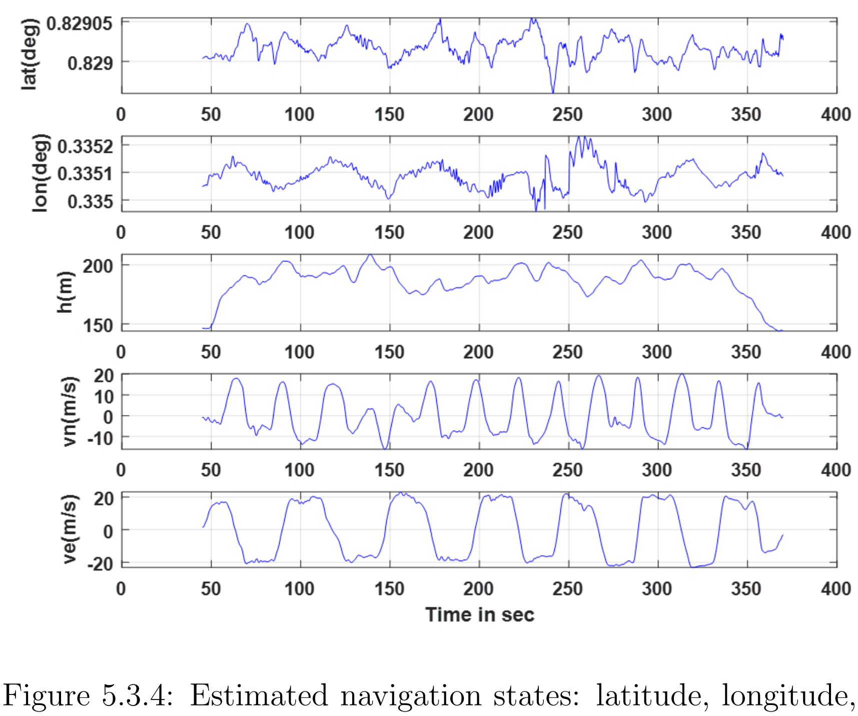

Figure 5.3.4: Estimated navigation states: latitude, longitude, altitude, North velocity, and East velocity

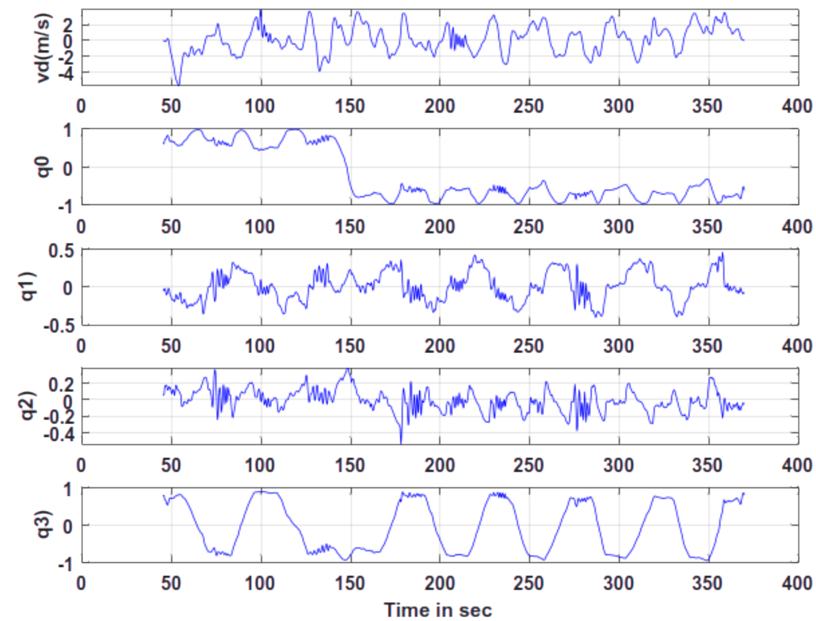

Figure 5.3.5: Estimated Down velocity and quaternion attitude states

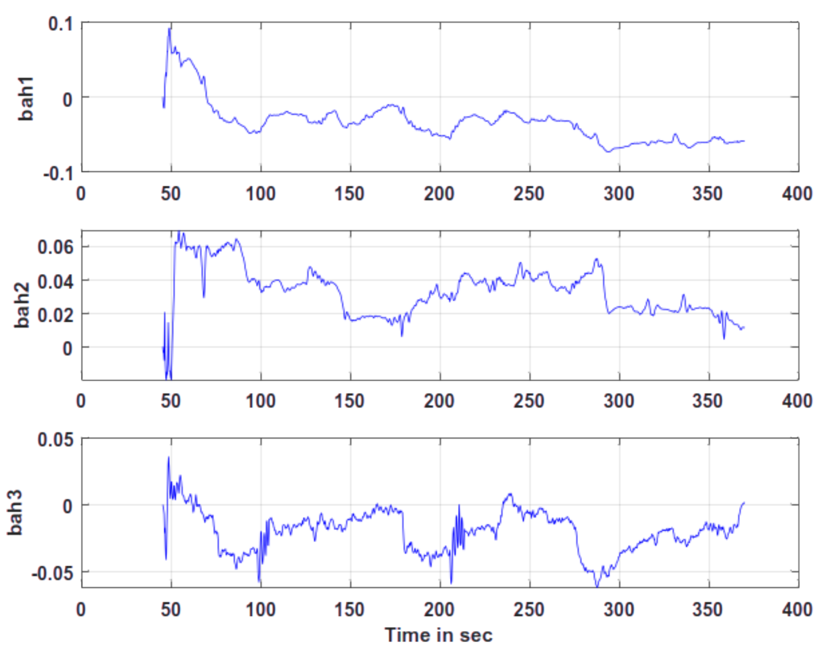

Figure 5.3.6: Estimated accelerometer bias states

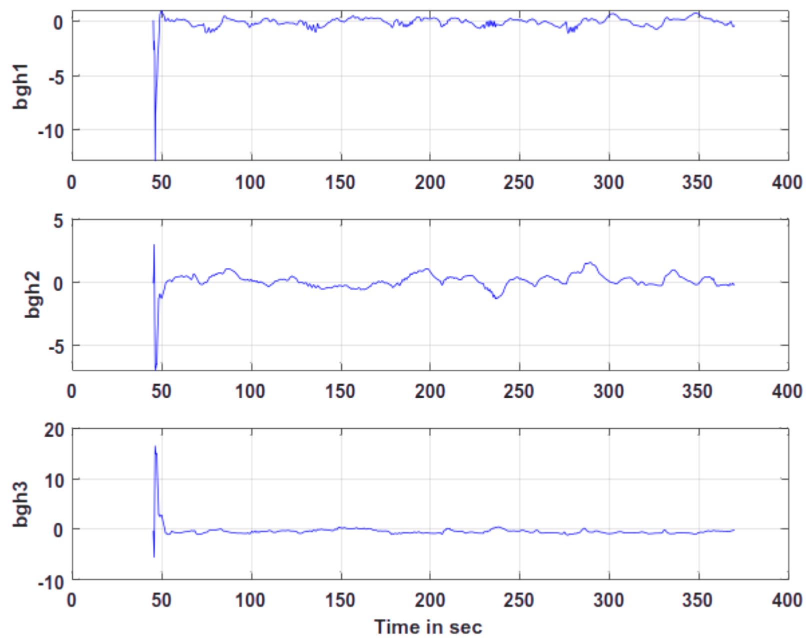

Figure 5.3.7: Estimated gyroscope bias states

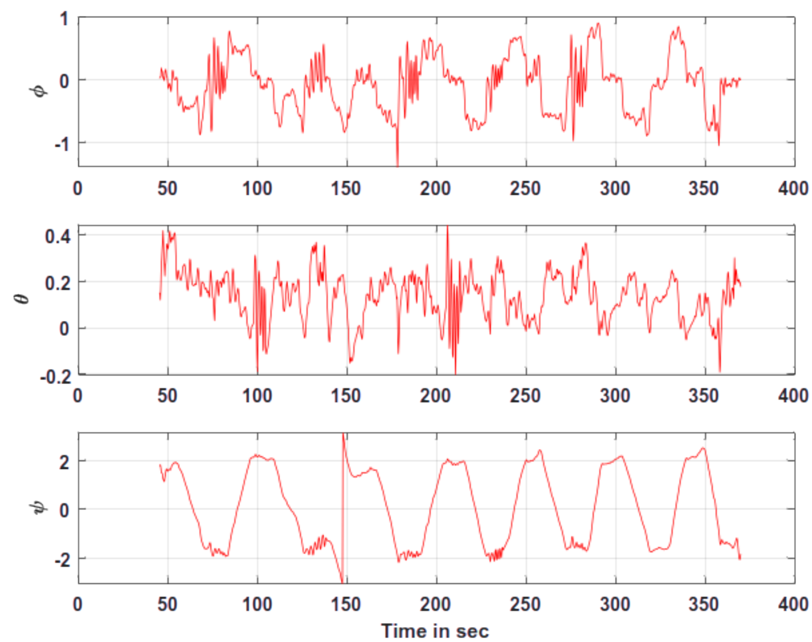

Figure 5.3.8: Estimated roll, pitch, and yaw attitudes

The incorporation of Neural Network-generated pseudo-GPS measurements significantly improves navigation performance during GPS outages. Compared with standalone INS operation, the proposed framework maintains bounded estimation errors and provides improved tracking of position, velocity, and attitude states. The results demonstrate that the Neural Network successfully learns the nonlinear INS drift characteristics and supplies useful correction information to the AUKF during GPS-denied operation.

5.4 Performance Comparison

This section evaluates the performance of the proposed AUKF–Neural Network navigation framework under GPS-denied conditions. The estimated navigation states are compared with GPS reference data, while the performance of different filtering approaches is analyzed using sensor bias estimation results. The objective is to assess the ability of the proposed framework to maintain navigation accuracy, reduce INS drift, and improve estimation robustness during GPS outages.

5.4.1 Comparison with GPS Reference Data

To validate the effectiveness of the proposed framework, the estimated navigation states obtained during GPS-denied operation are compared with the corresponding GPS reference measurements. This comparison provides a direct assessment of the framework's capability to reconstruct accurate position and velocity information in the absence of GPS updates.

Figure 5.4.1 and Figure 5.4.2 illustrate the comparison between the estimated latitude and longitude states and the corresponding GPS reference measurements. The estimated trajectories closely follow the reference data throughout the GPS outage period, indicating that the proposed AUKF–Neural Network framework is capable of effectively compensating for INS drift and maintaining accurate position estimation.

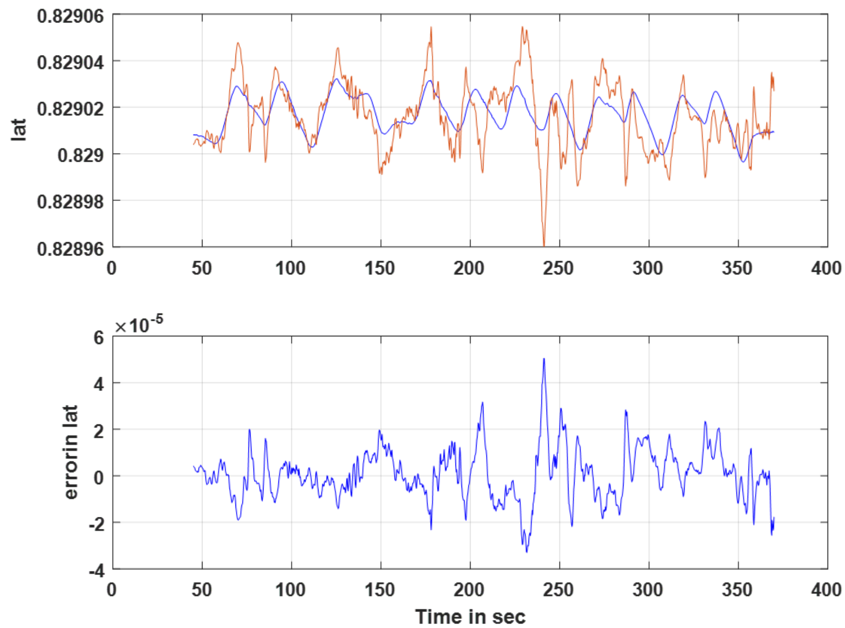

Figure 5.4.1: Comparison of latitude estimation during GPS-denied operation with GPS reference data

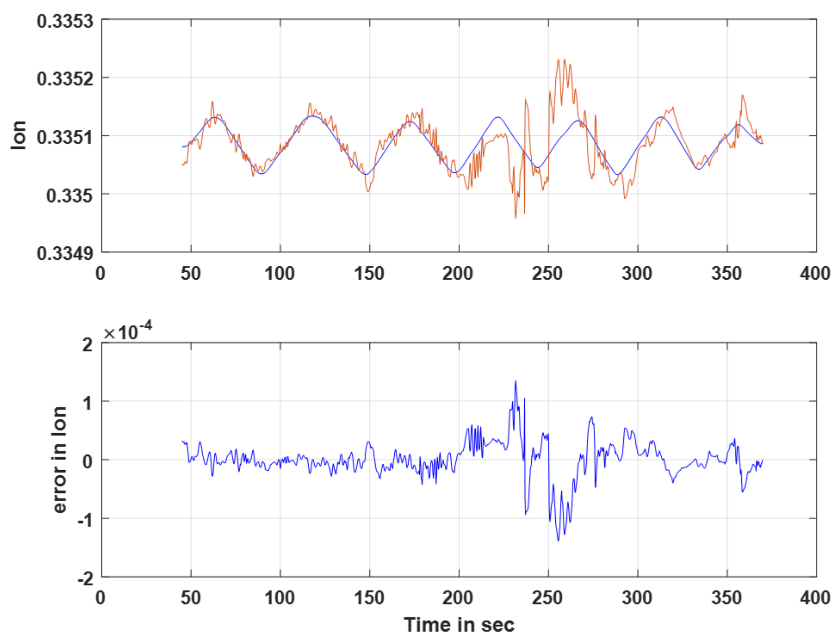

Figure 5.4.2: Comparison of longitude estimation during GPS-denied operation with GPS reference data

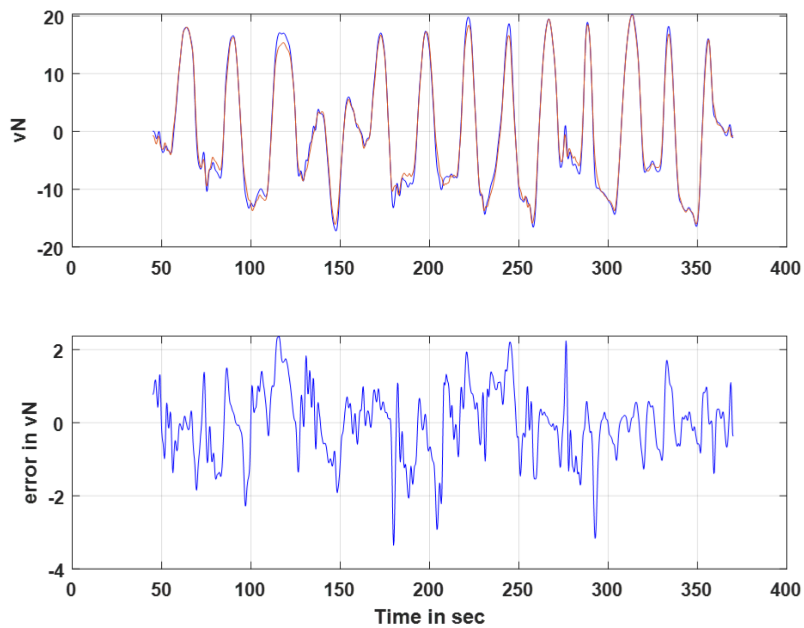

Figure 5.4.3: Comparison of north velocity estimation during GPS-denied operation with GPS reference data

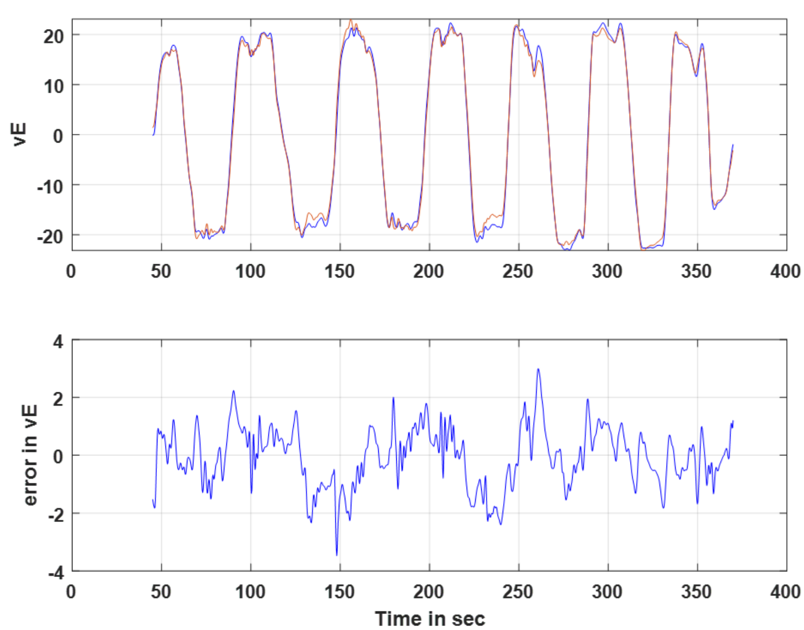

Figure 5.4.4: Comparison of east velocity estimation during GPS-denied operation with GPS reference data

Figure 5.4.3 and Figure 5.4.4 present the comparison between the estimated velocity states and GPS reference measurements. The results demonstrate that the proposed framework successfully preserves velocity accuracy and suppresses the error growth that normally occurs in standalone INS navigation. The close agreement between the estimated and reference velocity profiles validates the effectiveness of the Neural Network-generated pseudo-GPS corrections.

5.4.2 Navigation Function Validation

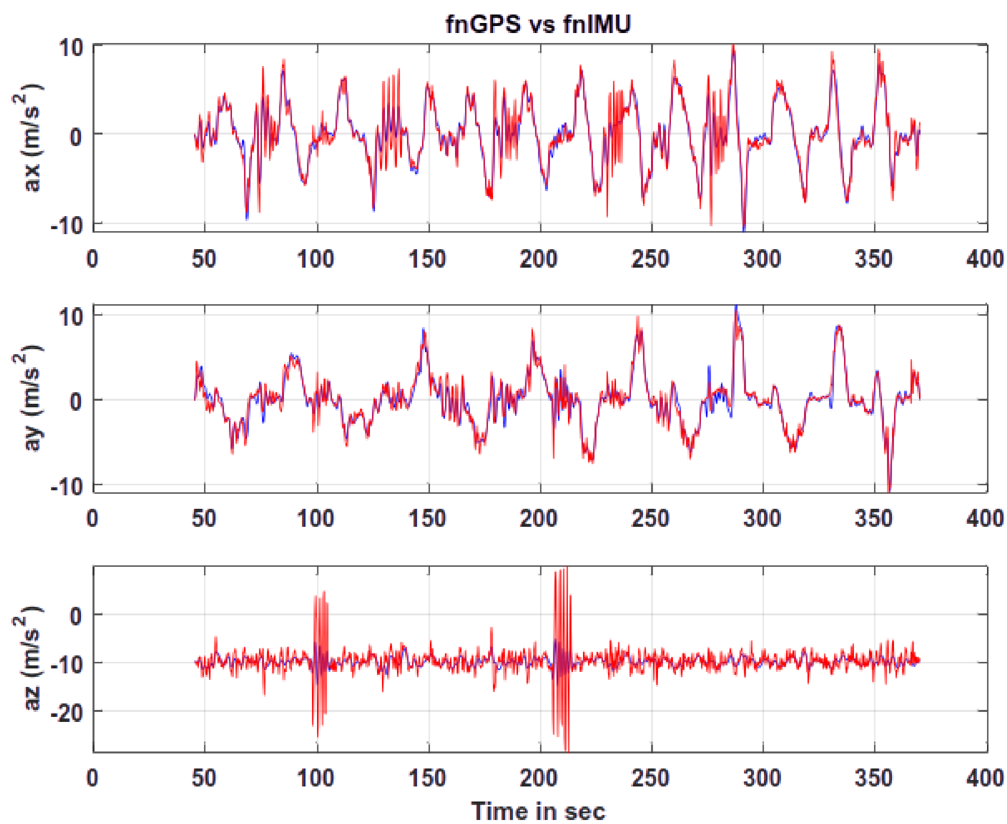

Figure 5.4.5: Comparison between GPS-derived and IMU-derived navigation function outputs

Figure 5.4.5 presents the comparison between the navigation function outputs computed using GPS measurements and those obtained from IMU data. The close agreement between the two signals indicates that the inertial measurements accurately capture the vehicle dynamics and can effectively be used for navigation state propagation. The result validates the consistency of the INS mechanization model and demonstrates that the inertial sensor measurements provide reliable information for the estimation process.

5.4.3 EKF and AUKF Performance Comparison

The performance of the Extended Kalman Filter (EKF) and Unscented Kalman Filter (UKF) is evaluated through the estimation of inertial sensor bias parameters. The estimated bias values and corresponding uncertainties are presented in Tables 5.4.1 and 5.4.2.

Table 5.4.1: Estimated sensor bias parameters using the Extended Kalman Filter

No	Bias	Parameter	Std. Deviation	Relative Std. Dev (%)
1	bah1	1.00000e-05	1.0019e-02	100193.99
2	bah2	1.00000e-05	1.0019e-02	100193.99
3	bah3	1.00000e-05	1.0019e-02	100193.99
4	bgh1	3.10417e-03	9.6222e-05	3.10
5	bgh2	1.40411e-04	7.9457e-05	56.59
6	bgh3	4.02195e-03	1.0759e-04	2.68

Table 5.4.2: Estimated sensor bias parameters using the Adaptive Unscented Kalman Filter

No	Bias	Parameter	Std. Deviation	Relative Std. Dev (%)
1	bah1	-5.04779e-02	1.9791e-04	0.39
2	bah2	1.51429e-02	2.1104e-04	1.39
3	bah3	1.39854e-02	1.7495e-04	1.25
4	bgh1	7.91035e-03	9.8276e-05	1.24
5	bgh2	2.44013e-03	7.9551e-05	3.26
6	bgh3	4.43710e-03	1.1274e-04	2.54

.

Tables 5.4.1 and 5.4.2 show that the UKF produces more consistent sensor bias estimates with significantly lower relative uncertainty than the EKF. Unlike the EKF, which relies on first-order linearization, the UKF propagates sigma points through the nonlinear system dynamics and captures higher-order nonlinear effects more accurately. Consequently, the UKF provides improved estimation performance and forms a more suitable foundation for the adaptive filtering framework adopted in this work.

Overall, the results demonstrate that the proposed AUKF–Neural Network framework effectively maintains navigation accuracy during GPS-denied operation. The generated pseudo-GPS measurements successfully compensate for INS drift, while the nonlinear filtering approach improves state estimation accuracy and robustness under uncertain operating conditions.

5.5 Chapter Summary

This chapter presented the simulation results and performance evaluation of the proposed AUKF–Neural Network based navigation framework for UAV operations in GPS-denied environments. The navigation performance was first evaluated under normal INS/GPS operation, followed by GPS-denied navigation using standalone INS propagation. The resulting degradation in estimation accuracy highlighted the limitations of conventional INS navigation during extended GPS outages.

To address this challenge, Neural Network-generated pseudo-GPS measurements were incorporated into the AUKF framework. The results demonstrated improved estimation accuracy, reduced position and velocity errors, and enhanced navigation continuity compared with standalone INS operation. Performance comparisons further confirmed the effectiveness of the proposed approach in compensating for INS drift and maintaining reliable UAV state estimation under uncertain operating conditions.

CHAPTER 6

Conclusions and Future Work

6.1 Conclusions

This work presented a hybrid UAV navigation framework for operation in GPS-denied environments by integrating the Adaptive Unscented Kalman Filter (AUKF) with a Neural Network (NN). An INS/GPS integrated navigation system was first developed using inertial sensor measurements and GPS observations. The nonlinear UAV dynamics were modeled in the North-East-Down (NED) frame, and quaternion-based attitude representation was employed to achieve stable and singularity-free attitude estimation.

To address the nonlinear characteristics of UAV navigation, the AUKF was implemented for state estimation and sensor fusion. While the AUKF provided accurate navigation performance during normal GPS availability, INS drift accumulated during GPS outages. To mitigate this issue, a Neural Network-based error compensation framework was developed. The Neural Network was trained using INS/GPS navigation data and was used to predict navigation errors and generate pseudo-GPS measurements during GPS-denied operation.

The results demonstrated that the proposed AUKF–Neural Network framework effectively reduced INS drift and maintained accurate position and velocity estimation during GPS outages. The generated pseudo-GPS measurements closely followed the GPS reference data, thereby improving navigation continuity and robustness. Overall, the proposed framework provides an effective and computationally efficient solution for UAV navigation in GPS-denied environments.

6.2 Key Contributions

The major contributions of this work are summarized as follows:

- Development of an INS/GPS integrated navigation framework for UAV state estimation.

- Implementation of the Adaptive Unscented Kalman Filter (AUKF) for nonlinear state estimation and sensor fusion.
- Development of a Neural Network-based approach for learning and compensating INS drift during GPS outages.
- Generation of pseudo-GPS measurements using Neural Network predictions to support navigation in GPS-denied environments.
- Validation of the proposed AUKF–Neural Network framework using UAV flight data and comparison with GPS reference measurements.

6.3 Future Work

Although the proposed AUKF–Neural Network framework demonstrated improved navigation performance during GPS-denied operation, several enhancements can be explored in future work.

- **Real-Time Implementation:**

  The developed framework can be implemented on an onboard UAV flight computer and evaluated under real-time operating conditions. This would enable assessment of computational requirements and practical deployment feasibility.

- **Advanced Neural Network Models:**

  More advanced architectures such as Long Short-Term Memory (LSTM) networks can be investigated to better capture the temporal characteristics of INS drift and improve prediction accuracy during extended GPS outages.

- **Long-Duration GPS Outages:**

  The proposed approach can be evaluated under longer GPS-denied intervals and more challenging flight scenarios to further assess its robustness and navigation accuracy.

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Publication Details